

UNIVERSITY OF SCIENCE, VNU-HCM

FACULTY OF INFORMATION TECHNOLOGY

TRAN THAI DANG KHOA

TELECOM STATION (POP) INVENTORY MANAGEMENT SYSTEM

(POP Management System)

GRADUATION INTERNSHIP PROJECT – BACHELOR OF INFORMATION TECHNOLOGY

REGULAR PROGRAM

Ho Chi Minh City, July 2022

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TRAN THAI DANG KHOA – 18120422

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SUPERVISOR

M.Sc. Pham Minh Tu

Ho Chi Minh City, July 2022

Acknowledgments

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Finally, I wish the faculty and the entire university leadership good health and continued success in their work.

I sincerely thank you!

Project Outline

Project title: Telecom Station (POP) Inventory Management System | (POP Management System) |

Ho Chi Minh City, April 04, 2022

Supervisor's confirmation Student

M.Sc. Pham Minh Tu Tran Thai Dang Khoa

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Abstract

Previously, whenever it wanted to view overview information or statistics on the number of incidents, the number of offline customers, POP status information, etc., the Southern Infrastructure Development and Management Center (INF MN) of FPT Telecom had to rely on a third-party piece of software, Telegram, using a chatbot to pull the needed data from the database. This was very time-consuming and inconvenient whenever a user simply wanted to review data that had already been retrieved before. In addition, the data was displayed in raw form and was not visual for the viewer.

Therefore, the company undertook the development of the Telecom Station (POP) Inventory Management System project. The focus of the project is to build an Overview Dashboard that visualizes management data, reports on POP information and status, tracks device configuration errors, tracks infrastructure-related incidents, and tracks the number of offline customers, among other things.

In this project, the student focused on researching web application design technologies and data visualization, and on building a complete application to manage telecom stations (POPs) – helping the company manage, track, report on, upgrade, and maintain telecom stations (POPs) in a faster and easier, automated way.

Chapter 1: Project Introduction

This chapter provides a general presentation of the project introduction, objectives, approach, and the structure of this internship project report.

General Introduction to the Project

The Telecom Station (POP) Inventory Management System project is a digitalization project of the Southern Infrastructure Development and Management Center (INF MN), under FPT Telecom. The project focuses on managing, tracking, reporting on, upgrading, and maintaining telecommunications stations (POPs). A POP is a telecom network equipment cabinet used to provide Internet service to customers. POPs are distributed widely across the country, concentrated mainly in major cities.

In this project, the student will study the company's systems and research the technologies needed to build an application for managing POPs. This includes designing an Overview Dashboard for POP information, the distribution of POPs by region, the number of incidents by region, incident resolution time, and so on.

Project Objectives

The project's objectives focus on system analysis, and on researching and surveying technologies for building the web application's Backend (NodeJS (Express), Python (Flask)) and Frontend (ReactJS (NextJS), VueJS, AngularJS), its database (Oracle, MS SQL Server, MongoDB), and data visualization along with supporting data-visualization libraries (Highcharts, ApexCharts) – then selecting the most suitable technologies to build the

system. This is followed by interface and system design to develop an application that meets the company's management, reporting, and infrastructure-tracking needs.

In addition, further features may be integrated to develop the application so that it meets a wider range of user requirements.

Project Approach

The project approach consists of the following steps:

- Study the system's business operations.
- Research relevant technologies and frameworks: overview documentation for these technologies and frameworks is available on their official websites.
- Analyze and design the system.
- Build the application: select the technologies and data visualization tools needed to build a complete application.
- Test, operate, and report on results.

Report Structure

The graduation internship project report consists of 6 chapters, summarized as follows:

- **Chapter 1 – Project Introduction:** Provides a general introduction to the project, including its general introduction, objectives, approach, and report structure.
- **Chapter 2 – Enterprise Business Analysis:** Presents the company's business operations, specifically the System Engineering Department. Gives a brief introduction to the company and the organization of the System Engineering Department, along with the difficulties the company is facing and possible solutions.
- **Chapter 3 – System Design Analysis:** System design analysis, business modeling, system requirement specification, use case diagram specification, and database design.
- **Chapter 4 – Technology Survey, Comparison, and Selection, and Data Visualization Research:** Presents an overview of the researched technologies, compares them, and selects the most suitable technologies for developing the system. Also covers research into data visualization and supporting libraries.
- **Chapter 5 – System Implementation, Data Visualization, and Evaluation:** Presents an overview of the system architecture, and provides a detailed presentation of the user interface design and the visualization of the system's data, along with an evaluation of the trial deployment results based on the criteria set out.

- **Chapter 6 – Conclusion and Future Directions:** Presents the results achieved, the difficulties encountered during implementation, an overall assessment, and proposed future directions for the application.

Chapter 2: Enterprise Business Analysis

This chapter presents the process of studying the company's business operations, specifically the operations of the System Engineering Department, the difficulties the company is facing, and possible solutions.

Overview of the Company – System Engineering Department

FPT Telecom Joint Stock Company (abbreviated as FPT Telecom) is currently one of the leading Telecommunications and Internet service providers in the region.

Founded on January 31, 1997, the company originated from the Online Service Center, founded by 4 members, along with Vietnam's first Internet network product named "Tri Tue Viet Nam – TTVN," a product regarded as having laid the foundation for the development of the Internet in Vietnam.

Business fields:

- Providing telecom network infrastructure for broadband Internet service
- Value-added services over the Internet and mobile networks
- Pay-TV service
- Messaging, data, and entertainment information services over mobile networks
- Building network infrastructure and providing Telecommunications and Internet services
- Import and export of Telecommunications and Internet equipment
- Local fixed-line Telecommunications service
- Value-added Telecommunications services
- Domestic long-distance fixed-line Telecommunications service
- Services for individual/mass-market customers:
- Fixed broadband Internet access service using FTTH/xPON technology
- FPT Play television, offering nearly 200 domestic and international TV channels, providing online learning applications, at-home fitness training, live-event applications, etc., allowing users to experience a wide range of needs – learning, entertainment, gaming, etc. – across multiple device platforms.
- Internet content and application services: FPT Play Box (Voice Remote), FPT Play, Foxy, Hi FPT, Fshare, Fsend, etc.

- IoT/Smart Home services and products: FPT Camera, iHome.
- Services for corporate and organizational customers:
- Data transmission: Domestic (local and inter-provincial connections) and international (IPLC, MPLS, IEPL).
- Dedicated Internet leased-line channels: NIX, GIA, Asia Transit.
- Voice services: Domestic (fixed-line telephone, VoIP, 1800/1900 hotlines) and international.
- Online data: Domain names, data and email hosting, server rental and colocation, Rack cabinet rental.
- Managed services: Video conferencing, cloud computing, systems integration, security services.
- Cloud computing service – FPT HI GIO CLOUD: a Cloud Infrastructure Service developed jointly by FPT Telecom and Internet Initiative Japan (IIJ). [1]

The System Engineering Department (KTHT) is part of the Southern Infrastructure Development and Management Center (INF MN) of FPT Telecom. It is responsible for building and managing access infrastructure stations and infrastructure devices from the MPOP level down, tracking configuration errors on telecom station (POP) equipment, tracking infrastructure-related incidents, resolving incidents, and tracking the number of offline customers, among other things.

The System Engineering Department is organized into several teams, each responsible for a different role:

- Operations and management team: tracks telecom station equipment information, plans infrastructure construction and management, and tracks incidents and affected customers. This team is further divided into 4 sub-teams, each managing one of 4 regions: Region 4, Region 5, Region 6, and Region 7.
- Incident handling and equipment maintenance team: receives incident and affected-customer information, resolves incidents, handles customer care, and performs equipment maintenance and upkeep.
- Telecom programming team: programs the telecom system, configures network devices, configures POP cabinets, sets up network paths, etc.
- Software development team: develops the telecom infrastructure management application and provides statistical data and infrastructure information to other departments.

Enterprise Business Analysis

Telecom Station System Management

To manage the telecom station system, the company needs to track overview infrastructure information and overview device-count statistics, and understand the utilization rate and distribution of telecom infrastructure by region in order to devise infrastructure construction and development strategies.

In addition, the company also needs to track statistics on infrastructure incidents and device errors, as well as information on affected customers, in order to devise optimal handling solutions and ensure the best experience for customers using telecom services.

The company's departments also produce periodic reports on the telecom station system by Month (Quarter, Half-Year, or Year). These include work-result reports, system-operation reports, and infrastructure statistics reports, used in meetings to review work progress and plan progress, and to propose new plans, new projects, and future directions for projects.

Telecom Equipment Management

Produces detailed statistics on the number and type of telecom equipment, its specifications, and its location by province and region. Periodically checks the condition and health of equipment, recording inspection information and assessments by severity level.

Produces statistics on device errors, assessing their severity as well as how frequently they occur on a device, in order to propose timely replacement or maintenance solutions.

Compiles and reports on telecom equipment information periodically by Month (Quarter, Half-Year, or Year).

Telecom Incident Management

Produces statistics on telecom incidents by type and detail, the cause of each incident, and incident resolution time by region, and devises handling solutions along with timely support.

Records information on affected and offline customers when an incident occurs, receives information from customers experiencing incidents, and quickly provides support and customer care to ensure a stable connection for the customer.

Produces periodic statistical reports on incidents by region and by time period (Week, Month, Quarter, Half-Year, or Year). Produces statistics on the number of affected customers and the duration of the impact, compares them, and devises optimal handling solutions.

Based on the results, incident resolution time, and support/customer care time, provides an assessment and comparison against the targets achieved.

Difficulties the Company Is Facing

When carrying out the business operations described above – for example, wanting to view overview information or statistics on the number of incidents, the number of offline customers, or POP status information – the company had to rely on a third-party piece of software, Telegram, using a chatbot to pull the needed data from the database.

From that information, the company would then compile the data needed to produce result reports and plan reports, broken down by region or by time period. This reporting work was also repeated on a Weekly, Monthly, Quarterly, Half-Yearly, or Yearly basis. This was very time-consuming and labor-intensive, since everything had to be done manually. Since the data came only from the chatbot, it was displayed only in raw form and was not visual for the viewer. In addition, it was inconvenient whenever the company simply wanted to review data that had already been retrieved before.

Therefore, the company proposed developing the Telecom Station (POP) Inventory Management System project. For this project, the student carried out the analysis, design, and construction of a system suited to data-reporting features, comparing statistical information by region and by time period, and, in particular, visualizing management data and reporting information in order to meet the company's business requirements.

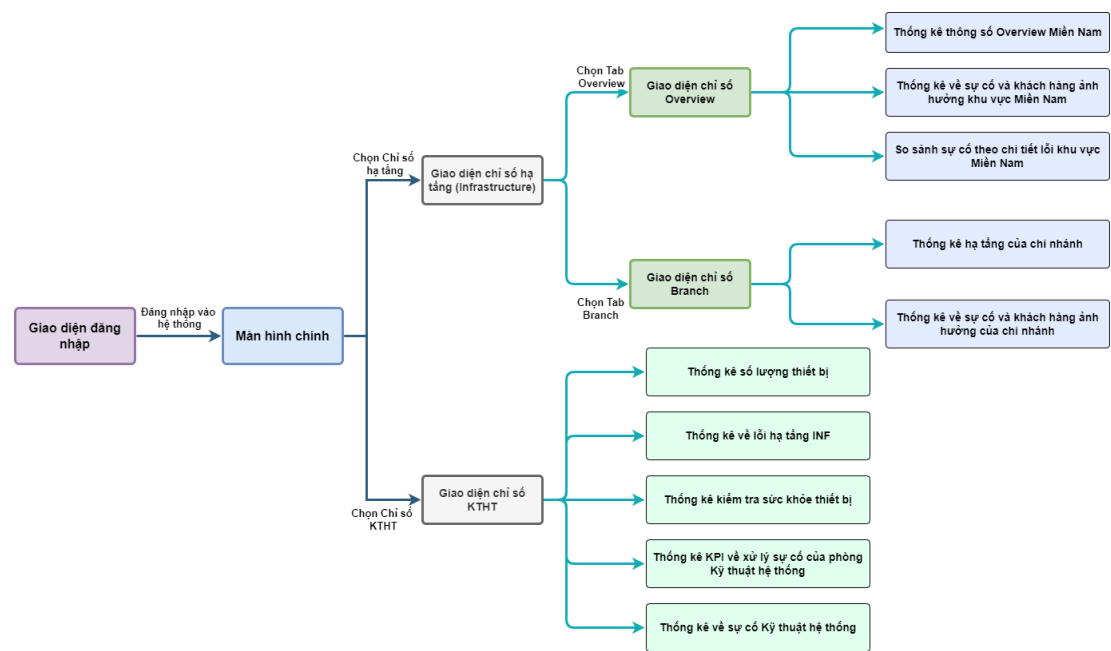
Chapter 3: System Design Analysis

This chapter presents, in detail, the analysis methodology, system requirement specification, business modeling, use case specification, and database design.

System Requirements Analysis and Specification

Based on the business analysis in the previous chapter, the student carried out the analysis and built the overview diagram of the telecom infrastructure management system.

System Overview Diagram



Functional Requirements

Function Code	Functional Requirement	Description
FR01	Login	The user can log in to the system. The login method is by username and password.
FR02	Southern Region Overview statistics	Overview statistics on infrastructure parameters, total device count, utilization rate, and infrastructure distribution for the Southern Region.
FR03	Southern Region incident and affected-customer statistics	Overview statistics on the number of incidents and the number of affected customers in the Southern Region.
FR04	Southern Region incident comparison by error detail	Compares the number of incidents and affected customers by error detail, over time periods, and by branch within the Southern Region.
FR05	Branch infrastructure statistics	Overview statistics on infrastructure parameters, total device count, and infrastructure utilization rate by branch.
FR06	Branch incident and affected-customer statistics	Statistics on the number of incidents and affected customers by error detail over time periods, for a given branch.
FR07	Device count statistics	Statistics on the number of devices managed by the System Engineering Department, by

Function Code	Functional Requirement	Description
		category.
FR08	INF infrastructure error statistics	Detailed statistics on device errors and incidents within the System Engineering Department.
FR09	Device health-check statistics	Statistics on device health-check information by severity level, and lists of devices with recorded physical or service errors.
FR10	System Engineering Department incident-handling KPI statistics	KPI statistics on incident resolution time and support handling time for the System Engineering Department.
FR11	System Engineering incident statistics	Statistics on incidents by type, detailed incident statistics, and support handling for the System Engineering Department.

Table 3.1: Functional requirements

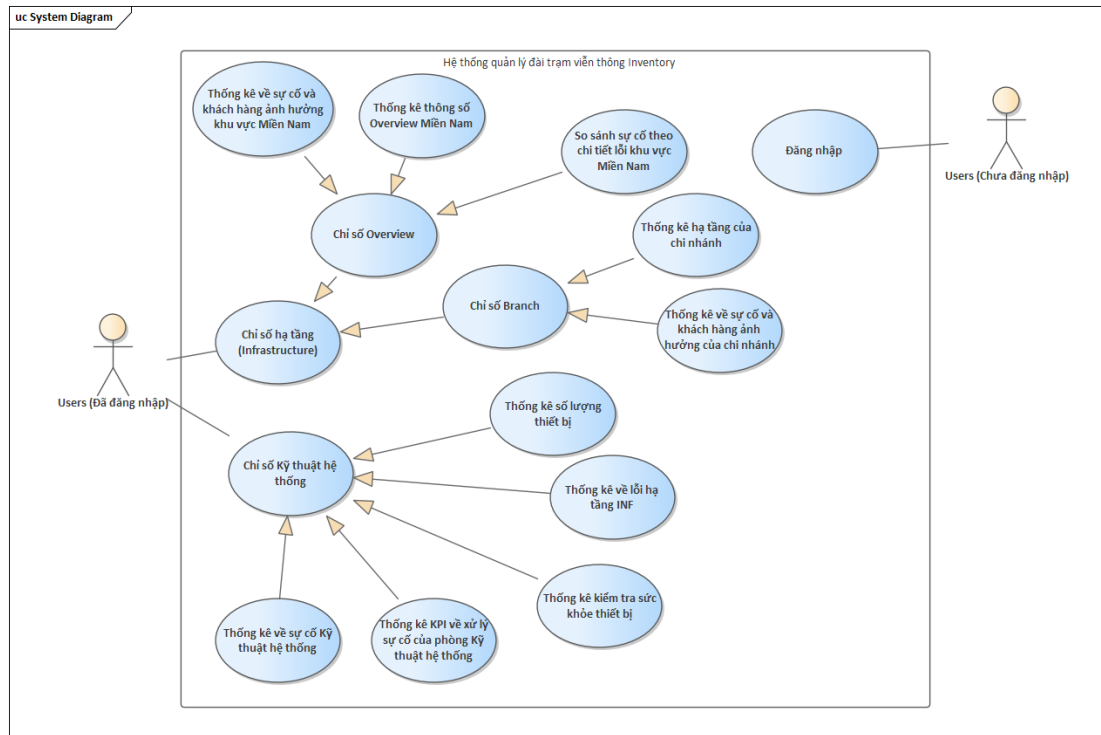
Non-Functional Requirements

Code	Category	Requirement
NonFunc01	Software platform	The system can run on different operating systems and most common browsers.
NonFunc02		The system is easy to maintain, improve, and extend (add new features).
NonFunc03	Performance	Fast page-load speed and fast data updates.
NonFunc04		The system operates 24/7.
NonFunc05	Security	High reliability; information and data in the system are kept secure.
NonFunc06	Interface	Reasonable layout, easy to look at, harmonious colors.
NonFunc07	User experience	Simple operation, easy to use for all types of users.

Table 3.2: Non-functional requirements

Business Modeling and System Use Case Specification

Use Case Diagram



Use Case Specification

Field	Description
Use case name	Login
Description	The user must enter their information (username/password) to log in to the system.
Actor	User (not yet logged in)
Priority	High / Mandatory
Trigger	–
Preconditions	The system already has account information.
Postconditions	The system is in the logged-in state.
Main flow	1. The system displays the user login screen. 2. The user enters their account information (username and password). 3. The user clicks the login button. 4. The system validates the login information. 5. The system navigates to the application's home screen once the login information is valid.
Alternate flow	A4. If the username and password are invalid, the system displays a message to the user and asks them to re-enter their credentials to log in.

Table 3.3: Use case specification – FR01

Field	Description
Use case name	Southern Region Overview statistics
Description	Displays a screen with overview statistics on infrastructure parameters, total device count, utilization rate, and infrastructure distribution for the Southern Region.
Actor	User (logged in)
Priority	High / Mandatory
Trigger	After the user selects the Infrastructure Metrics section.
Preconditions	–
Postconditions	–
Main flow	1. The system displays overview statistics on Southern Region parameters. 2. The user can interact visually with the data charts.
Alternate flow	A2. In the infrastructure-distribution statistics, the user can view statistics filtered by specific criteria by selecting the criteria to view.

Table 3.4: Use case specification – FR02

Field	Description
Use case name	Southern Region incident and affected-customer statistics
Description	Displays overview statistics on the number of incidents and the number of affected customers in the Southern Region.
Actor	User (logged in)
Priority	High / Mandatory
Trigger	After the user selects the Infrastructure Metrics section.
Preconditions	–
Postconditions	–
Main flow	1. The system displays statistics on incidents and affected customers in the Southern Region. 2. The user can interact visually with the data charts.
Alternate flow	A2. The user can choose to view data filtered by Week/Month/Quarter/Half-Year/Year.

Table 3.5: Use case specification – FR03

Field	Description
Use case name	Southern Region incident comparison by error detail
Description	Displays a chart comparing the number of incidents and affected customers by error detail, over time periods, and by branch within the Southern Region.
Actor	User (logged in)
Priority	High / Mandatory

Field	Description
Trigger	After the user selects the Infrastructure Metrics section.
Preconditions	–
Postconditions	–
Main flow	1. Displays comparison charts of incident counts by error detail within the Southern Region. 2. The user can interact visually with the data charts.
Alternate flow	A2. The user can select a time period or an error detail to filter the data as desired.

Table 3.6: Use case specification – FR04

Field	Description
Use case name	Branch infrastructure statistics
Description	Displays an overview statistics screen for infrastructure parameters, total device count, and infrastructure utilization rate by branch.
Actor	User (logged in)
Priority	High / Mandatory
Trigger	After the user selects the Infrastructure Metrics section and the Branch tab.
Preconditions	–
Postconditions	–
Main flow	1. Displays overview statistics on infrastructure metrics, total device count, and infrastructure utilization rate by branch. 2. The user can select any statistics item shown in the system to view its detailed information.
Alternate flow	A2. When the user selects any item, the screen displays a detailed list of information for that item.

Table 3.7: Use case specification – FR05

Field	Description
Use case name	Branch incident and affected-customer statistics
Description	Displays statistics on the number of incidents and affected customers by error detail, over time periods, for a branch.
Actor	User (logged in)
Priority	High / Mandatory
Trigger	After the user selects the Infrastructure Metrics section and the Branch tab.
Preconditions	–
Postconditions	–
Main flow	1. Displays charts of incident counts by error detail for the branch,

Field	Description
	compared across time periods and error details. 2. The user can interact visually with the data charts.
Alternate flow	A2. The user can choose to view data filtered by Week/Month/Quarter/Half-Year/Year, by a specific time period, or by error detail.

Table 3.8: Use case specification – FR06

Field	Description
Use case name	Device count statistics
Description	Displays a screen with statistics on the number of devices managed by the System Engineering Department, by category.
Actor	User (logged in)
Priority	High / Mandatory
Trigger	After the user selects the System Engineering Metrics section.
Preconditions	The user must be a staff member of the System Engineering Department, or a person with authority above that department.
Postconditions	–
Main flow	1. Displays a chart of device counts managed by the System Engineering Department over time periods. 2. The user can interact visually with the data charts.
Alternate flow	A2. The user can choose to display the chart by Total device count/Device type/Function/Model.

Table 3.9: Use case specification – FR07

Field	Description
Use case name	INF infrastructure error statistics
Description	Displays detailed statistics on device errors and incidents within the System Engineering Department.
Actor	User (logged in)
Priority	High / Mandatory
Trigger	After the user selects the System Engineering Metrics section.
Preconditions	The user must be a staff member of the System Engineering Department, or a person with authority above that department.
Postconditions	–
Main flow	1. Displays detailed statistics on device errors and incidents within the System Engineering Department. 2. The user can select optional fields to display data as desired.
Alternate flow	A2. The user can filter data by region, province, or branch.

Table 3.10: Use case specification – FR08

Field	Description
Use case name	Device health-check statistics
Description	Displays a screen with statistics on device health-check information by severity level, and a list of devices with recorded physical or service errors.
Actor	User (logged in)
Priority	High / Mandatory
Trigger	After the user selects the System Engineering Metrics section.
Preconditions	The user must be a staff member of the System Engineering Department, or a person with authority above that department.
Postconditions	–
Main flow	1. Displays charts of device health-check information, and a list of devices with recorded physical and service errors. 2. The user can interact visually with the data charts.
Alternate flow	A2. The user can filter chart data by region, province, branch, device type, device name, and record severity level.

Table 3.11: Use case specification – FR09

Field	Description
Use case name	System Engineering Department incident-handling KPI statistics
Description	Displays KPI statistics on incident resolution time and support handling time for the System Engineering Department.
Actor	User (logged in)
Priority	High / Mandatory
Trigger	After the user selects the System Engineering Metrics section.
Preconditions	The user must be a staff member of the System Engineering Department, or a person with authority above that department.
Postconditions	–
Main flow	1. Displays KPI statistics on incident resolution time and support handling time for the System Engineering Department. 2. The user can interact visually with the data charts.
Alternate flow	A2. The user can choose to display the chart by Month/Quarter/Half-Year/Year.

Table 3.12: Use case specification – FR10

Field	Description
Use case name	System Engineering incident statistics

Field	Description
Description	Displays incident statistics by type, detailed incident statistics, and support handling for the System Engineering Department.
Actor	User (logged in)
Priority	High / Mandatory
Trigger	After the user selects the System Engineering Metrics section.
Preconditions	The user must be a staff member of the System Engineering Department, or a person with authority above that department.
Postconditions	–
Main flow	1. Displays incident statistics by type, detailed incident statistics, and support handling for the System Engineering Department. 2. The user can interact visually with the data charts.
Alternate flow	A2. The user can choose to display the chart by Month/Quarter/Half-Year/Year.

Table 3.13: Use case specification – FR11

Database Design Analysis

Because the company's database had already been designed long ago and stores a very large volume of data, the project reuses the company's existing data. The student only modified a few parts to fit the project's needs, while keeping the database's original structure unchanged, since re-engineering it would have significantly affected the system and been fairly costly.

Database Model

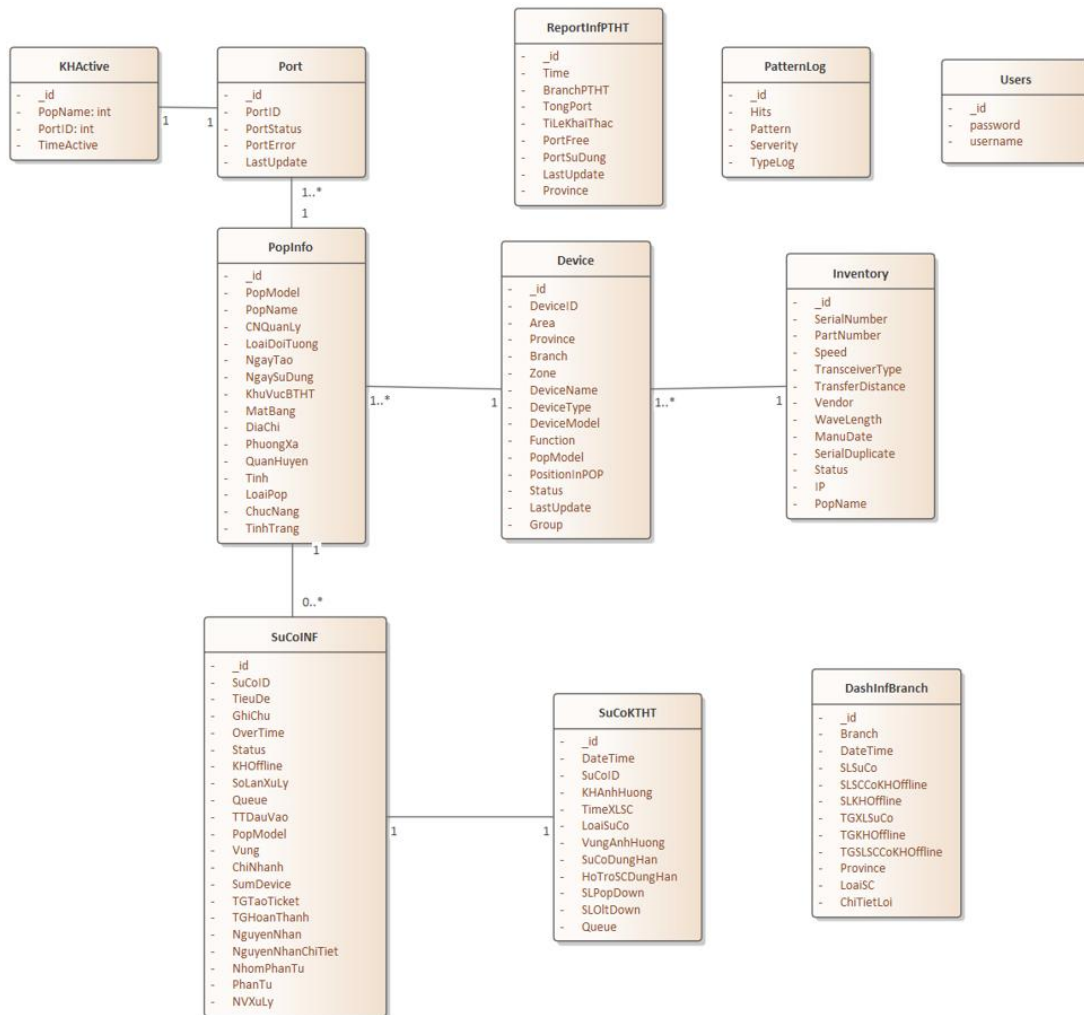


Figure 3.3: Database model

Database Design

PopInfo – table storing POP information

Attribute Name	Data Type	Description
_id	ObjectId	PopInfo identifier
PopModel	String	POP model
PopName	String	POP name
CNQuanLy	String	Managing branch information
LoaiDoiTuong	String	POP object type
NgayTao	Datetime	POP creation date
NgaySuDung	Datetime	POP in-service date

Attribute Name	Data Type	Description
KhuVucBTHT	String	System maintenance area information
MatBang	String	Site/premises information
DiaChi	String	POP address
PhuongXa	String	Ward/Commune
QuanHuyen	String	District
Tinh	String	Province
LoaiPop	String	POP type
ChucNang	String	POP function
TinhTrang	String	POP status

Table 3.14: Database – PopInfo table

Device – table storing device information

Attribute Name	Data Type	Description
_id	ObjectId	Device identifier
DeviceID	String	Device code
Area	String	Area information
Province	String	Province
Branch	String	Branch
Zone	String	Zone information
DeviceName	String	Device name
DeviceType	String	Device type
DeviceModel	String	Device model
Function	String	Function
PopModel	String	POP model
PositionInPop	String	Position within the POP
Status	String	Status
LastUpdate	Datetime	Last update time
Group	String	Device group

Table 3.15: Database – Device table

Inventory – table storing device inventory information

Attribute Name	Data Type	Description
_id	ObjectId	Inventory identifier
SerialNumber	String	Device serial number
PartNumber	String	Part number

Attribute Name	Data Type	Description
Speed	String	Device speed
TransceiverType	String	Transceiver type
TransferDistance	String	Transfer distance
Vendor	String	Vendor
WaveLength	String	Wavelength
ManuDate	String	Manufacture date
SerialDuplicate	String	Duplicate serial number
Status	String	Status
IP	String	IP address
PopName	String	POP name

Table 3.16: Database – Inventory table

SuCoINF – table storing Southern Region infrastructure incident information

Attribute Name	Data Type	Description
_id	ObjectId	SuCoINF identifier
SuCoID	String	Incident code
TieuDe	String	Incident title
GhiChu	String	Incident notes
OverTime	String	Overdue incident flag
Status	String	Incident status
KHOffline	Int	Number of offline customers
SoLanXuLy	Int	Number of handling attempts
Queue	Int	Incident queue
TTDauVao	String	Input information
PopModel	String	POP model
Vung	String	Region information
ChiNhanh	String	Branch information
SumDevice	Int	Total number of affected devices
TGTaoTicket	Datetime	Ticket creation time
TGHoanThanh	Datetime	Incident resolution completion time
NguyenNhan	String	Incident cause
NguyenNhanChiTiet	String	Detailed incident cause
NhomPhanTu	String	Affected element group
PhanTu	String	Affected element
NVXuLy	String	Handling staff member information

Table 3.17: Database – SuCoINF table

SuCoKTHT – table storing System Engineering Department incident information

Attribute Name	Data Type	Description
_id	ObjectId	SuCoKTHT identifier
DateTime	Datetime	Time the incident occurred
SuCoID	String	Incident code
KHAnhHuong	Int	Number of affected customers
TimeXLSC	Int	Incident resolution time
LoaiSuCo	String	Incident type
VungAnhHuong	String	Affected region
SuCoDungHan	Int	Number of incidents resolved on time
HoTroSCDungHan	Int	Number of support cases handled on time
SLPopDown	Int	Number of POP-down events
SLOLTDown	Int	Number of OLT-down events
Queue	Int	Queue position

Table 3.18: Database – SuCoKTHT table

DashInfBranch – table storing Southern Region branch dashboard information

Attribute Name	Data Type	Description
_id	ObjectId	DashInfBranch identifier
Branch	String	Branch
DateTime	Int	Time period
LSuCo	Int	Number of incidents
SLSCCoKHOOffline	Int	Number of incidents with offline customers
SLKHOOffline	Int	Number of offline customers
TGXLSuCo	Int	Incident resolution time
TGKHOOffline	Int	Customer offline duration
TGXLSCKHOOffline	Int	Resolution time for incidents with offline customers
Province	String	Province
LoaiSC	String	Incident type
ChiTietLoi	String	Error detail

Table 3.19: Database – DashInfBranch table

ReportInfPTHT – table storing Southern Region infrastructure development report information

Table 3.20: Database – ReportInfPTHT table

Attribute Name	Data Type	Description
_id	ObjectId	ReportInfPTHT identifier
Time	Datetime	Time period
BranchPTHT	String	Branch information
TongPort	Int	Total number of ports
TiLeKhaiThac	Double	Infrastructure utilization rate
PortFree	Int	Number of free ports
PortSuDung	Int	Number of ports in use
LastUpdate	Datetime	Last update time
Province	String	Province

PatternLog – table storing device health-check record information

Attribute Name	Data Type	Description
_id	ObjectId	PatternLog identifier
Hits	Int	Number of error occurrences
Pattern	String	Record pattern
Serverity	String	Severity level
TypeLog	String	Record type

Table 3.21: Database – PatternLog table

Users – table storing user information

Attribute Name	Data Type	Description
_id	ObjectId	Users identifier
Username	String	Login username
Password	String	Password

Table 3.22: Database – Users table

Port – table storing port information

Attribute Name	Data Type	Description
_id	ObjectId	Port identifier
PortID	String	Port code
PortStatus	String	Port status
PortError	String	Port error information
LastUpdate	Datetime	Last update time

Table 3.23: Database – Port table

KHActive – table storing active-customer information

Attribute Name	Data Type	Description
_id	ObjectId	KHActive identifier
PopName	String	POP name
PortID	String	Port code
TimeActive	Datetime	Activation time

Table 3.24: Database – KHActive table

Chapter 4: Technology Survey, Comparison and Selection, and Data Visualization Research

This chapter presents a survey of the technologies used to build the web application, compares them, and selects the technologies best suited to building a system that meets the stated requirements. It also covers data visualization research and supporting libraries.

Database

Oracle

Oracle is a database management system, whose full name is Oracle Database, or OracleDB. In fact, “Oracle” is also the name of Oracle Corporation, the publisher of OracleDB. Oracle is built on SQL, a standardized programming language used by database administrators, data analysts, and data professionals to manage and query stored data.



Oracle logo

Figure 4.1: Oracle

Oracle supports large databases and both numeric and character data types, while also providing storage support for object-oriented databases. The Oracle database can manage multiple databases using a two-phase commit protocol. Oracle supports cursors, making programming easier. It also executes each transaction separately, and the result of each transaction is invisible to other transactions until it completes, which increases data security. [2]

Oracle Database offers the following features to meet database management requirements:

- **Scalability and high performance:** features such as Real Application Clustering and Portability make Oracle Database highly scalable depending on usage. Oracle provides performance-enhancing solutions such as Oracle Advanced Compression and Oracle Database In-Memory to optimize the system's operating performance as much as possible.
- **Availability:** to support database availability, Oracle provides the Oracle Data Guard feature. When using this feature, a secondary standby database is maintained as a copy of the primary database and can be used as an alternative during failover.
- **Backup and Recovery:** Oracle has fully developed Recovery features to restore data after most types of failure. In a failure scenario, the database needs to be recovered immediately to ensure high availability. Unaffected portions of data remain available while the affected portions are still being recovered.
- **Security:** information security is continuously given top priority. Oracle provides mechanisms to control access to and use of data. Enforcing authorization and regulating client behavior can restrict unlawful access while still allowing individual user access.

Microsoft SQL Server

Microsoft SQL Server is a popular database management system developed by Microsoft, based on the RDBMS (Relational Database Management System) model.

SQL Server provides a full set of management tools, from a GUI interface to a query language for SQL. SQL's strength lies in the number of platforms it can be combined with, such as ASP.NET and C# for building Winform applications, while it can also operate independently. However, SQL Server



typically ships with its own implementations of the SQL and T-SQL languages, among others.

SQL Server is optimized to run in Very Large Database Environments, up to the terabyte scale, and can serve thousands of users at once. SQL Server integrates well with other servers such as Microsoft Internet Information Server (IIS), E-Commerce Server, Proxy Server, and others. [3]

Microsoft SQL Server has a wide range of features, the most notable of which are:

- **Specialized enterprise features:** Microsoft now offers a wide variety of data management features, business intelligence (BI) features, and SQL Server analytics tools. Alongside the Machine Learning Services first integrated in SQL Server 2016, the data analysis services include SQL Server Analysis Services – a data processing and analysis tool used in BI – data visualization applications, and SQL Server Reporting Services, which supports creating and distributing BI reports.
- **Management features:** on the administration side, Microsoft SQL Server includes SQL Server Integration Services, SQL Server Data Quality Services, and SQL Server Master Data Services. There are also two toolsets dedicated to DBAs and developers: SQL Server Data Tools, used for database development, and SQL Server Management Studio, used to deploy, monitor, and manage databases.

MongoDB

MongoDB is an open-source database first developed in 2007. It is a leading NoSQL database used by millions of people, and it is written in C++.



MongoDB is also a cross-platform database that operates around the concepts of Collections and Documents. It offers high performance, high availability, and easy scalability.

Strengths of NoSQL:

- NoSQL solves big-data problems for information systems or for distributed data.
- Flexible scaling: NoSQL replaces the old mantra of database administrators, “scale up,” with a new one: “scale out.” Instead of adding bigger servers to handle more data, a NoSQL database lets a company distribute the load across many servers as the load grows.
- High availability: MongoDB’s automatic replication capability makes it very robust, because in the event of any failure, the data is automatically replicated back to its previous consistent state.

Commonly used MongoDB terminology:

- _id – A required field present in every document. The _id field represents a unique value within a MongoDB document, and can also be understood as the document’s

primary key. If you add a new document, MongoDB automatically generates an `_id` for it that is unique across the MongoDB database.

- **Collection** – A group of multiple documents in MongoDB. A Collection can be understood as roughly equivalent to a table in an RDBMS (Relational Database Management System). A collection resides within a single database. Collections do not require columns, rows, or data types to be defined in advance.
- **Cursor** – A pointer to the result set of a query. A client can iterate over a cursor to retrieve the results.
- **Database** – The container that holds Collections, similar to how an RDBMS database holds tables. Each database has its own file stored on physical storage. A single MongoDB server can hold multiple databases.
- **Document** – A single record within a Collection is called a Document. Documents in turn consist of field names and values.
- **Field** – A name-value pair within a document. A document can have zero or many fields. Fields are similar to columns in a relational database.
- **JSON** – Short for JavaScript Object Notation. A human-readable, simple text format representing structured data. JSON is currently supported by a great many programming languages.
- **Index** – A special data structure used to hold a small portion of a dataset in a form that is easy to scan. An index stores the value of a specific field or set of fields, sorted by the values of those fields. Indexes support efficient query resolution. Without an index, MongoDB has to scan every document in a collection to select the documents that match the query. This scan is inefficient and requires MongoDB to process a large volume of data [4].

The Aggregation Framework in MongoDB

MongoDB offers many powerful data-processing features, but the most prominent is Aggregation. This is a very useful feature for computing over large volumes of data across a set of documents (similar to a table in SQL).

Some basic Aggregation operations:

- `$project`: specifies the desired fields to query.
- `$match`: selects the desired documents to query.
- `$limit`: limits the number of documents.
- `$skip`: skips a certain number of documents.
- `$group`: groups documents by a given condition.

- \$sort: sorts documents.
- \$unwind: performs an expansion operation on an array, producing one output document for each value in that array.
- \$out: writes the result of the pipeline to a collection. [5]

Comparison between SQL and the Aggregation framework

SQL Command	Aggregation framework operator
SELECT	\$project, \$group, \$function: \$sum, \$min, \$avg, ...
FROM	db.collectionName.aggregate(...)
JOIN	\$unwind
GROUP BY	\$group
HAVING	\$match

Table 4.1: Comparison between SQL and the Aggregation framework

Comparison and Selection

Comparison:

Criterion	Oracle	Microsoft SQL Server	MongoDB
HIGH COMPLEXITY	Yes	No	No
STABILITY	Yes	Yes	Yes
HIGH PERFORMANCE	Yes	Yes	Yes
REUSABILITY	Yes	Yes	Yes
BIG-DATA PROCESSING	Yes	Yes	No
SECURITY	Yes	Yes	Yes
QUERY SPEED	Yes	Yes	Yes
CROSS-PLATFORM	Yes	Yes	Yes
FLEXIBILITY	No	No	Yes
HORIZONTAL DATA SCALABILITY	No	No	Yes
HIGH OPERATING COST	Yes	Yes	No

Table 4.2: Comparison of database management systems

Selection:

Based on the table above comparing the strengths and weaknesses of these database management systems, MongoDB can be selected as the system's primary database, because its flexibility and horizontal scalability suit the storage of device information and incident information – these two tables change frequently, and the number of data fields in each also varies by type.

MongoDB also integrates the Aggregation framework, a feature that supports data retrieval and greatly simplifies data processing for developers.

In addition, since the company only wants to build a system that is as simple and optimal as possible, Oracle and Microsoft SQL Server are not good choices, because both of these products carry fairly high licensing costs.

Selected: MongoDB

Backend Technologies

Python (Flask)

Python is a high-level, open-source programming language, first released in 1991 by Guido van Rossum.



Python programming language

Figure 4.4: Python

Python supports server-side web development, including the complex backend functions that websites carry out to display information to users. For example, websites need to interact with a database, communicate with other websites, and protect data as it travels across the network.

Python is useful for server-side programming because the language provides many libraries containing ready-made code for complex backend functions. Developers also use a range of Python frameworks that provide all the tools needed to build web applications quickly and easily. For example, developers can scaffold a framework-based web application in the blink of an eye because they don't need to build it from scratch. They can then test this web application using the framework's own testing tools, without depending on external testing tools. [6]



Flask is a micro-framework written in Python, used to develop small web applications. This framework's characteristics include enthusiastic community support, useful documentation, scaffolding tools, unit-testing features, and a built-in web server. It also provides extensions to support authentication, database mapping layers, and web security.

Notable features of Flask:

- Built-in development server
- RESTful request dispatching
- Secure cookie support
- Full WSGI compliance
- Compatibility with Google App Engine tooling
- Numerous extensions for desired features
- Built-in HTTP request handling
- A unique and coherent API

Express NodeJS



Express.js, or simply Express, is a backend web framework built on top of Node.js, released as free, open-source software for all users. It provides powerful features for developing web applications and APIs [7]. Express.js supports HTTP methods and middleware, making it extremely powerful and easy to use for building APIs.

Express.js only requires the JavaScript programming language, which makes building web applications and APIs simpler for newcomers to web application development.

With Express.js, users don't need to go through many complex programming steps to build an efficient API; Express also makes routing easier and provides many ready-made templates to choose from for development.

Notable features of Express.js:

- **Fast server development:** Express.js provides many features in the form of functions that can easily be used anywhere in the program. This removes the need to write code from scratch, saving time.
- **Middleware:** middleware software that has access to the database, the client's request, and other middleware. This middleware is mainly responsible for the systematic organization of Express.js's functions.
- **Routing:** Express.js provides a routing mechanism that helps maintain the state of the website with the help of URLs.
- **Templating:** the templating tools provided by Express.js allow developers to build dynamic content on websites by creating HTML templates on the server side.
- **Debugging:** debugging is indispensable to successfully developing web applications. With Express.js, debugging has become easier thanks to the ability to pinpoint exactly which parts of the web application have errors. [8]

Comparison and Selection

Comparison:

Criterion	Python	Express NodeJS
OPEN SOURCE	Yes	Yes
HIGH PERFORMANCE	No	Yes
RESOURCE REUSE	Yes	Yes
SUPPORTING LIBRARIES	Yes	Yes
SECURITY	Yes	Yes
REST API	Yes	Yes
SUPPORTS MVC-PATTERN APPLICATION DEVELOPMENT	Yes	Yes
MULTI-THREADED	No	Yes
SINGLE-THREADED	Yes	Yes
HIGH EXECUTION SPEED	No	Yes
SCALABILITY	No	Yes
CROSS-PLATFORM	Yes	Yes

Table 4.3: Comparison of Backend technologies

Selection:

Looking at the table comparing the strengths and weaknesses of Python and Express NodeJS, we can see that these two technologies are fairly similar for building a Backend web application.

According to the company's requirements, the system needs to have high execution speed and performance, and it also needs to be scalable and upgradable to support long-term development.

Therefore, choosing Express NodeJS is the appropriate choice for designing the system's Backend.

Selected: Express NodeJS

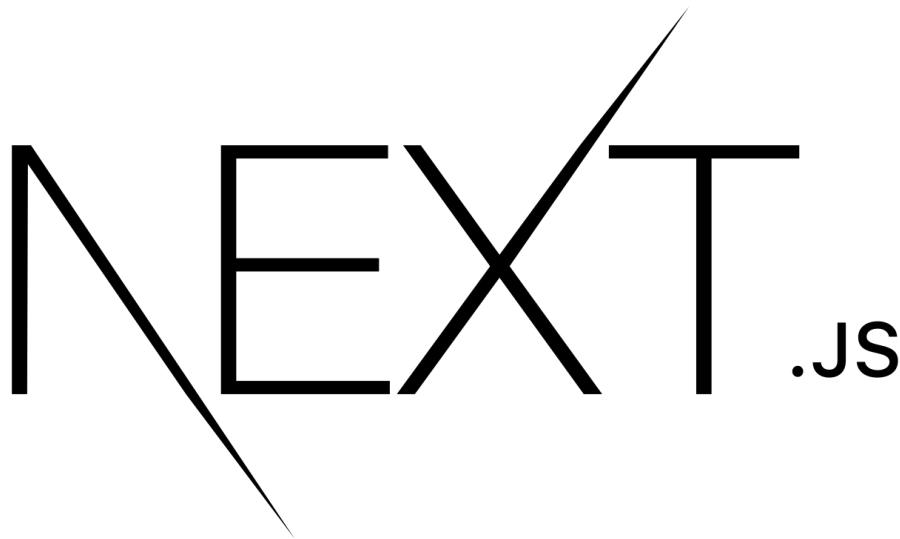
Frontend Technologies

ReactJS (NextJS)

React (also known as React.js or ReactJS) is a free, open-source JavaScript frontend library for building component-based user interfaces. It is maintained by Meta (formerly Facebook) along with a community of individual developers and businesses [9].



ReactJS not only makes websites feel smoother, it also greatly increases execution speed. ReactJS is highly scalable and simple for newcomers to learn. In React, instead of using plain JavaScript to lay out a page, developers use JSX. JSX is considered simpler to use than JavaScript and allows HTML to be quoted directly, using HTML tag syntax to render subcomponents. JSX is optimized at compile time, so it runs faster than the equivalent JavaScript code.



React is often used as the foundation for building web or mobile applications together with frameworks such as NextJS.

NextJS is a compact framework that makes it easy to build Single Page App / Server-Side Rendering applications with ReactJS.

NextJS integrates many features, such as:

- Excellent React support with outstanding performance.
- Fast feature development without spending too much time on configuration, such as Static, Webpack, and Babel setup.
- Pre-rendering, including both static generation (SSG) and server-side rendering (SSR).
- Automatic code splitting for faster page loads – each page only loads what it needs. For example, when rendering the home page, other pages are not initialized, so the home page loads quickly even if your website has 100 pages.

- Support for CSS and Sass, and support for any CSS or JS library.
- Fast page refresh in the development environment.
- Fully extensible.

AngularJS



AngularJS is a JavaScript web framework developed and sponsored by Google. It is a structured framework that supports building dynamic web applications. Angular is open source, completely free, and developed by thousands of skilled developers around the world.

AngularJS is often used for frontend development, calling data through APIs, and uses a powerful MVC model.

Core features of AngularJS:

- Data-binding: automatically synchronizes data between the model and the view.
- Scope: the binding between the Controller and the View.
- Controller: JavaScript functions that handle logic in conjunction with the Scope.
- Service: as mentioned above, AngularJS mainly uses APIs built from web services (PHP, ASP) to interact with the database.
- Filters: select elements from an array and return a new array.
- Directives: mark up DOM elements, meaning they create custom HTML tags.
- Templates: information displayed from the controller; a component of views.
- Routing: creates transitions between actions in the controller.
- MVC: a model that separates components into Model, View, and Controller. This is considered a fairly good model, though in Angular it is “adapted” somewhat to be closer to MVVM.

- Deep Linking: allows you to encode the state of the application in URLs so it can be bookmarked with search engines.
- Dependency Injection: Angular makes it easier for developers to build, understand, and test applications.

VueJS



Vue.js is a JavaScript framework created by Evan You that helps build Frontends as well as Single Page Applications. Vue was designed from the ground up to allow and encourage incremental application development. When building the interface layer, users only need to use Vue's core library, which is very easy to learn and integrates well with existing libraries or projects.

In essence, Vue.js uses an MVC (Model-View-Controller) architecture and can be used together with different architectures, such as CBA (Component-Based Architecture). Its ability to interoperate with many different existing frameworks has made Vue very popular in the web development ecosystem. Applications built with VueJS also have very fast page-load and processing speed.

Comparison and Selection

Comparison:

Criterion	ReactJS (NextJS)	AngularJS	VueJS
SUPPORTS FAST, EFFICIENT UI DEVELOPMENT	Yes	Yes	Yes
SINGLE PAGE APPLICATION	Yes	Yes	Yes
TWO-WAY BINDING	No	Yes	Yes
COMPREHENSIVE DOCUMENTATION	Yes	No	Yes
LARGE DEVELOPMENT COMMUNITY	Yes	Yes	No
SUPPORTING LIBRARIES AND TOOLS	Yes	Yes	Yes

Criterion	ReactJS (NextJS)	AngularJS	VueJS
EASY TO LEARN, EASY TO USE	Yes	No	Yes
SUITABLE FOR LARGE-SCALE PROJECTS	Yes	Yes	No
AUTOMATED ROUTER / WEBPACK	Yes	No	No
FLEXIBILITY	Yes	No	No
STABILITY	Yes	Yes	Yes
HIGH PERFORMANCE	Yes	Yes	Yes
REUSABILITY	Yes	Yes	Yes
READY-MADE PACKAGES	Yes	No	No
LONG-TERM DEVELOPMENT	Yes	Yes	No

Table 4.4: Comparison of Frontend technologies

Selection:

All three technologies above are fairly similar in terms of the features needed to design a web Frontend. ReactJS with NextJS is relatively more stable and more popular than VueJS, and also has a larger community and more documentation. NextJS's automatic Route and Webpack handling makes building the Frontend easier and saves a lot of time. AngularJS has a clear architecture and is backed by Google, but learning AngularJS is more difficult and demanding.

ReactJS leans toward building applications for the long term, with many ready-made packages available. JSX gives developers the full power of JavaScript, and with its large community, its long-term development prospects are better.

Based on these factors, ReactJS with NextJS can be selected to design the system's Frontend, thanks to the conveniences and automated features this framework supports.

Selected: ReactJS (NextJS)

Data Visualization Research

Data Visualization

What Is Data Visualization?

Data visualization is the graphical representation of information and data. By using visual elements such as charts, graphs, and maps, data visualization tools provide an accessible way for viewers and readers to understand what is being communicated, grasp information, and get the most out of the data.

In the world of Big Data, data visualization tools and technologies are essential for analyzing large volumes of information and making data-driven decisions.

Data visualization is also of great help in analysis and in producing overview reports on data, as well as in comparing pieces of data against one another.

Different chart and graph types in data visualization:

- Comparison: when we want to know how research subjects differ from one another.
- Composition: when we want to know specifically what components make up something – a group, etc. – and what proportion each component represents.
- Distribution: when we want to examine the distribution of data, its spread, its concentration, or to identify anomalies or outliers in the data.
- Trend: when we want to understand the trend of change for a research subject within the data.
- Relationship/Correlation: when we want to understand the relationship or correlation between two or more research subjects within the data.

Ways of Using Data Visualization

a) Change over time

This is the most basic and most common way of using data visualization. The reason it's the most common is that most data is related to time in some way. Therefore, the first step in being able to analyze such data is to look at what the trend of the visualized data looks like over time.

b) Determining frequency

Frequency is a fairly basic use of data visualization, since it, too, applies to data related to time.

c) Identifying relationships (correlations)

Identifying correlations is an extremely valuable use of data visualization. It is very difficult to identify the relationship between two variables without data visualization, yet it is important to be aware of relationships within the data. This is a great example of the value data visualization brings to data analysis.

d) Examining a network

An example of examining a network with data visualization can be seen in market research. Marketing professionals need to know how their brand is perceived by their target audience, so they can analyze the entire market and identify audience groups, or segment the market, find bridges between groups, identify influencers within each group, and understand other things around their customer base.

e) Planning

When planning or scheduling a complex project, everything can get confusing. Gantt charts solve this problem by clearly illustrating each task within a project and how long it takes to complete.

f) Value and risk analysis

Identifying complex metrics such as value and risk requires many different variables to be brought together, which makes it very hard to see clearly in a simple spreadsheet. It can be as simple as color-coding based on a formula to indicate which opportunities carry which values and corresponding risks [13].

Data Visualization Tools and Selection

Microsoft Power BI



Microsoft Power BI is a data analysis and visualization tool that lets you extract information from many data sources and bring it into interactive dashboards. It is an ideal tool for companies that want real-time reporting on their most important metrics and KPIs.

This tool is well-suited for gathering complex data and turning it into easy-to-understand charts and infographics. Power BI is also useful for companies collecting large volumes of data from many sources. Power BI's tools can easily combine everything into one comprehensive report.

Power BI is cloud-based, so many team members can access and share data across multiple Microsoft Office applications.

Tableau



Tableau is an analytics platform that is very good at compiling data from many different fields, such as algorithms, artificial intelligence, and business intelligence. Their data visualization software is also known for being user-friendly, which makes it easy for many companies to adopt.

Tableau integrates fully with Salesforce, so any business using that CRM benefits from this tool. It's one of 250 applications that businesses can use with it.

For businesses with a smaller budget, Tableau Public is offered as a free service for creating interactive data visualizations online.

Sisense

Sisense is a data analytics and business-intelligence software that lets non-technical people participate in, analyze, and visualize large datasets within an interactive dashboard. Sisense also lets users customize their data visualizations and embed them almost anywhere. The software analyzes user information and then presents it in a more visual and interactive way so that everyone has a chance to take part in exploring the data.



Sisense logo

Figure 4.13: Sisense

The tool's open cloud platform lets it collect information from many different sources and lets many people access this data. It also offers the ability to integrate with many other Sisense products.

If users want to share interactive visuals, this tool lets them publish them in many different formats, including PDF, PPT, Excel, Word, and others.

JavaScript Data Visualization Tools

JavaScript has a great many different libraries that developers can use to solve virtually any real-world problem – including data analysis and visualization. Highcharts and ApexCharts are two prime examples of chart-based data visualization libraries, thanks to how easy they are to use and how complete their documentation is for newcomers.

a) Highcharts



Highcharts is a JavaScript library that helps developers visualize data and create graphs and charts, letting users publish charts directly on a website. Highcharts can meet almost every requirement for charting, as well as for interactive charts in web applications. Highcharts offers many chart types, for example line charts, spline charts, area charts, bar charts, pie charts, and so on.

Because Highcharts is built in JavaScript, it can be easily integrated into websites and interacts well with many programming languages.



b) ApexCharts

ApexCharts is a modern, free, open-source charting library that helps developers create beautiful, interactive visuals for websites. It has built-in support for three of today's popular JavaScript libraries/frameworks: Angular, React, and Vue.

Selection

Because the system's data visualization needs are only at a basic visual level and do not require heavy data analysis, the student chose to use the simplest and easiest-to-use tools to design the data visualizations. Since the system is built on the JavaScript programming ecosystem (ReactJS, NodeJS), choosing Highcharts and ApexCharts is the most suitable and optimal choice for the system.

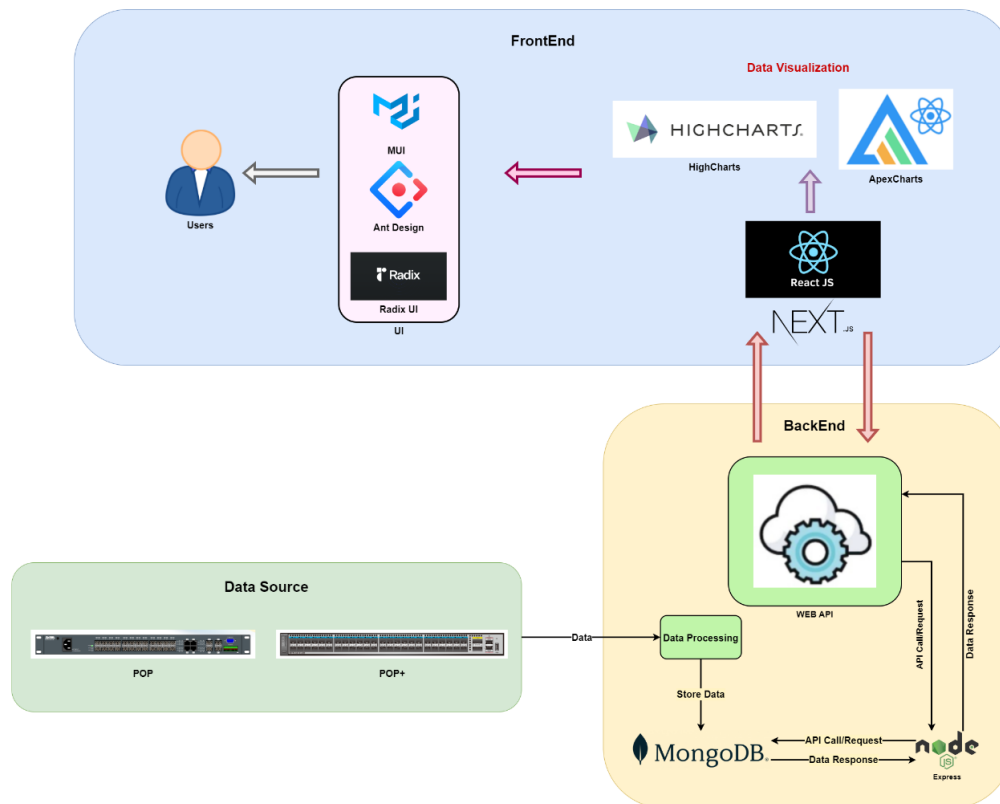
Selected: Highcharts and ApexCharts

Chapter 5: System Implementation, Data Visualization, and Evaluation

This chapter presents, in detail, the process of building the application and the main features implemented. It walks through the user interface screens and how data is visualized in charts. Finally, it presents an evaluation of the results of the application's trial deployment.

System Implementation and Data Visualization

System Architecture



System architecture

Data is recorded from the POPs/POP+s and sent for processing. It is then stored in MongoDB. From MongoDB, Web APIs are built with Express NodeJS on the Backend.

When a user loads the web page or performs an action such as viewing statistics or filtering data by time, status, and so on, the system receives the request from the Client and sends it down to the Server, which calls the API Request. The Server sends the request and receives the response data back from the Database. The data is then converted into the appropriate Web APIs and sent back to the Client. The Client then visualizes the received API data and displays it on the user interface as tables and charts, so users can view system information as well as the statistics they need.

System Components

After the analysis, design, and technology research phases, the student built the Infrastructure Management Web Dashboard following the architecture above, with the following components:

- A Web Dashboard application that visualizes data using the Highcharts and ApexCharts libraries, making it easy and intuitive to report on and manage the system.
- A MongoDB database with the Aggregation framework, which offers many powerful features, fast query speed, and horizontal scalability, giving the system more flexible data organization.
- A server stack made up of NextJS and Express.js, integrating many frameworks and supporting libraries that make the system easier to build, greatly increase execution speed, and offer high scalability, allowing the system to be upgraded and developed over the long term.

Interface and Data Visualization

Login Screen

To use the system, users must log in through the login page. Users enter the login credentials they were given (username and password) and click Login to enter the system.

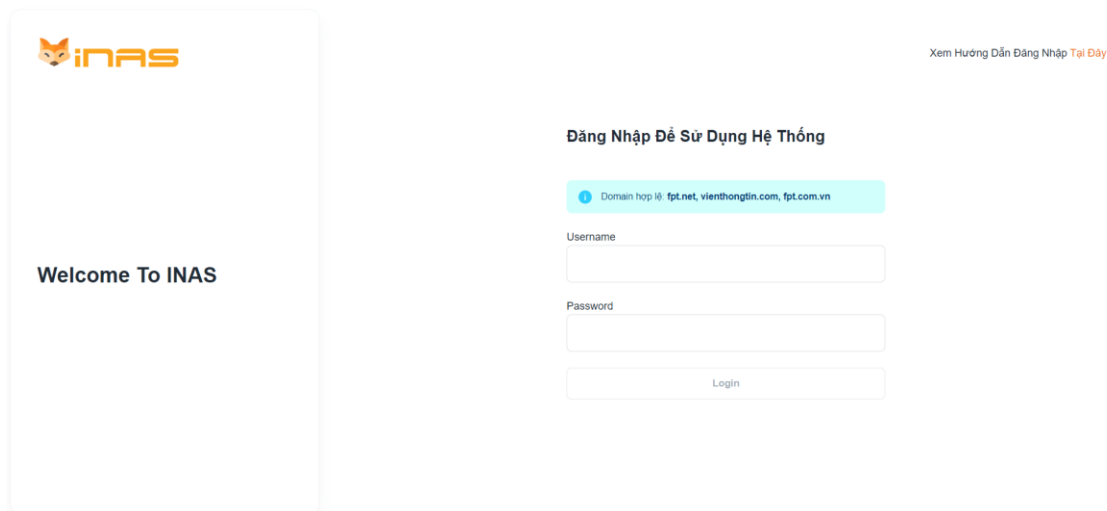


Figure 5.2: Login screen

Home Screen

After logging in and having their credentials validated by the system, the user is taken to the home screen. Here, the user can choose between two options: viewing the Southern Region's Infrastructure Metrics, or the System Engineering Department's Metrics.

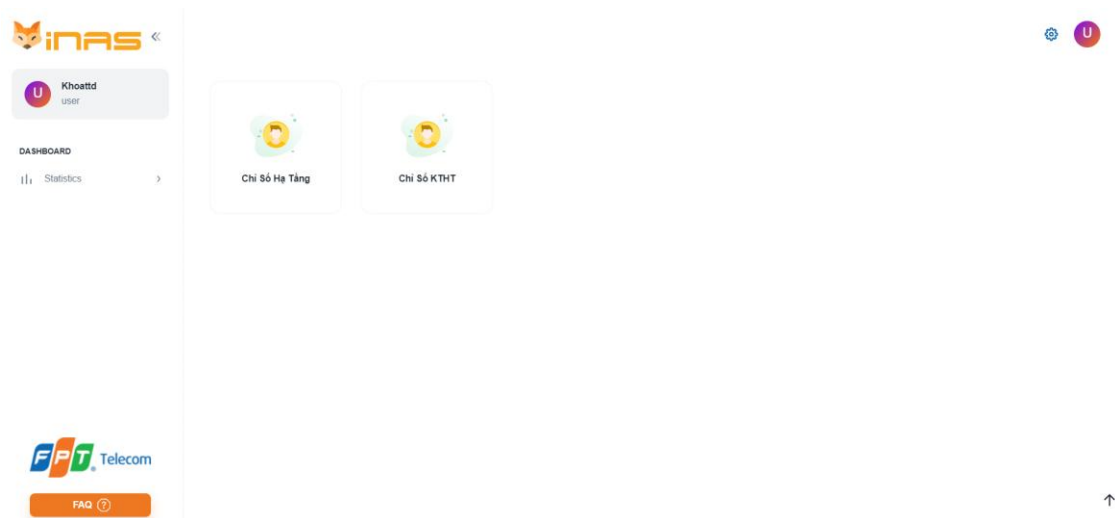


Figure 5.3: Home screen

Dashboard Overview Page

When the user selects Infrastructure Metrics, the system navigates to the Southern Region Infrastructure Management Overview Dashboard page. This screen displays the reporting features and the options available for each feature.

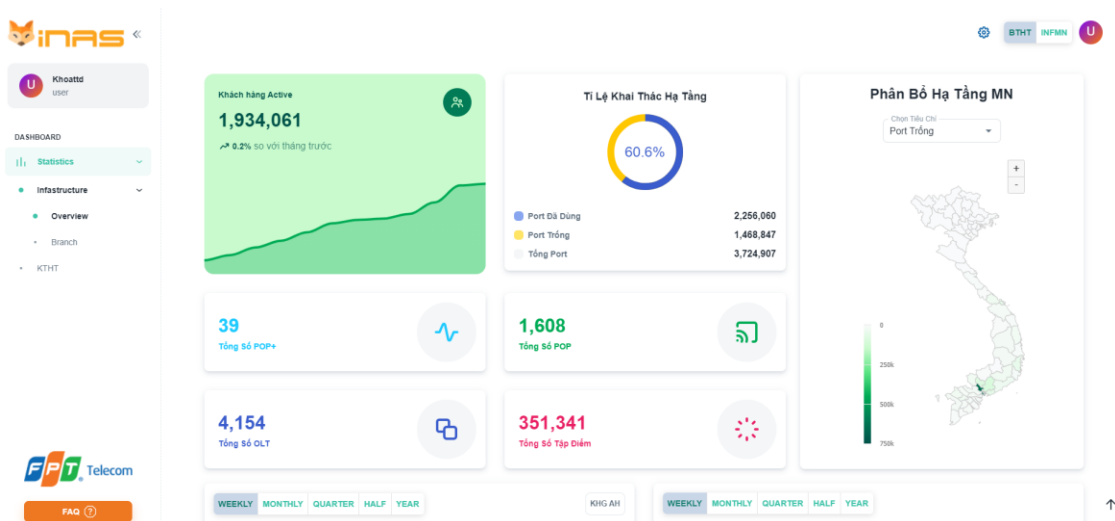


Figure 5.4: Dashboard Overview page

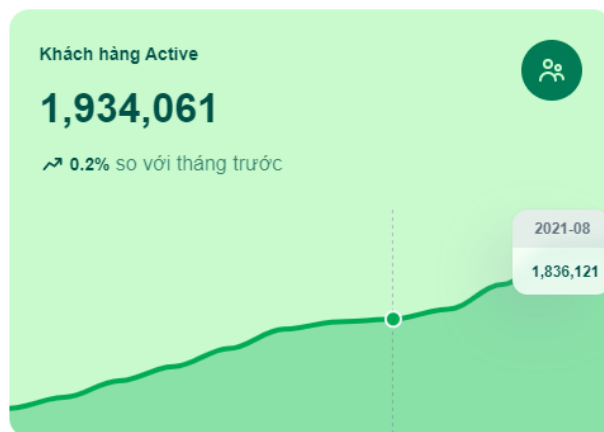
Southern Region Overview Statistics

Below are the charts that visualize the Southern Region infrastructure overview parameters, including: the number of Active Customers over the last 12 months, the infrastructure utilization rate, the infrastructure distribution (which can be filtered by specific criteria), and device statistics.

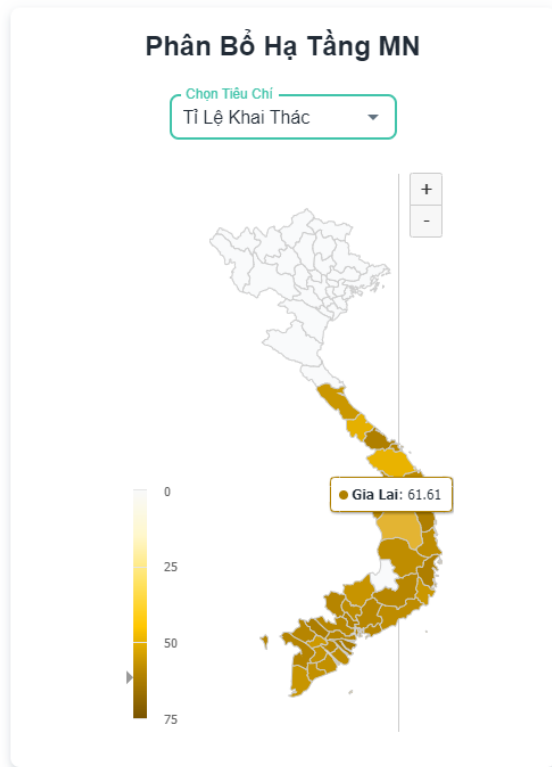


Figure 5.5: Southern Region overview statistics

As shown on the screen above, the infrastructure utilization rate is visualized as a pie chart that clearly shows the split between the number of ports in use and the number of free ports, out of the total number of ports across the whole region. Below the chart, there are also statistics for each component of the chart.



The chart above is a line chart showing the number of Active Customers in the system over the last 12 months, showing the specific figures, the current total, and the growth rate compared to the previous month. Based on this chart, we can see how the number of Active Customers rises and falls month to month.



This chart shows the infrastructure distribution as a map, using the HighchartMap feature in Highcharts to visualize the data, broken down by province across the Southern Region, with the option to view infrastructure distribution statistics filtered by criteria.

Southern Region Incident and Affected-Customer Statistics

The next feature on the Overview page is overview statistics on the number of incidents, the number of offline customers, the average incident resolution time, and the average customer offline duration for the whole Southern Region over time, with the option to view statistics by Week, Month, Quarter, Half-Year, or Year.

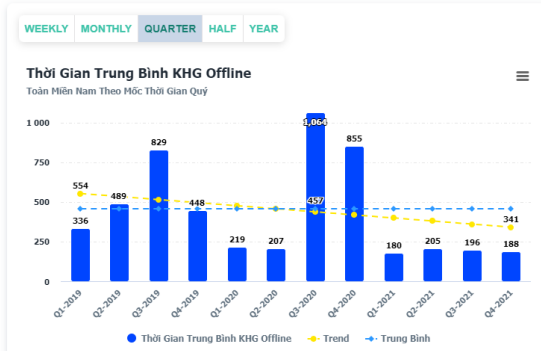
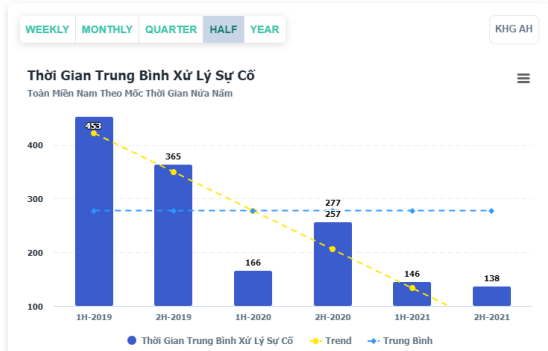
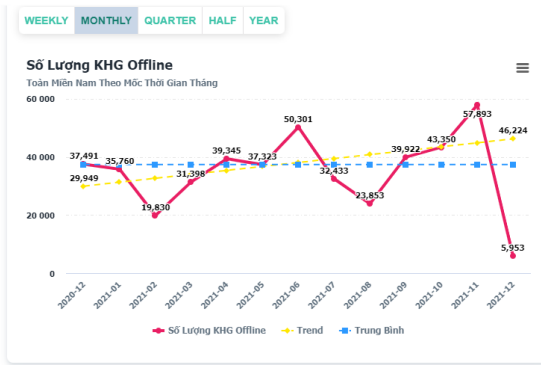
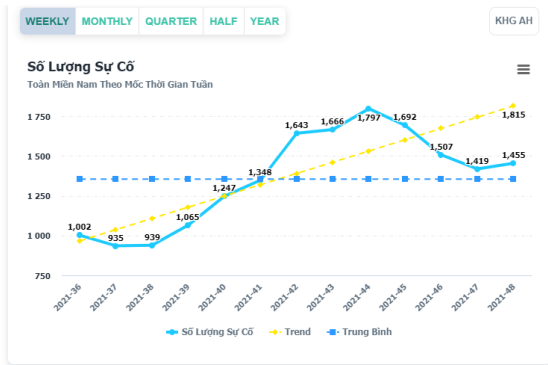


Figure 5.8: Incident and affected-customer statistics for the Southern Region

These charts are visualized using line and bar charts, clearly showing the rise and fall in both counts and average times. The charts also include lines showing the upward/downward trend as well as the moving average of the data over time.

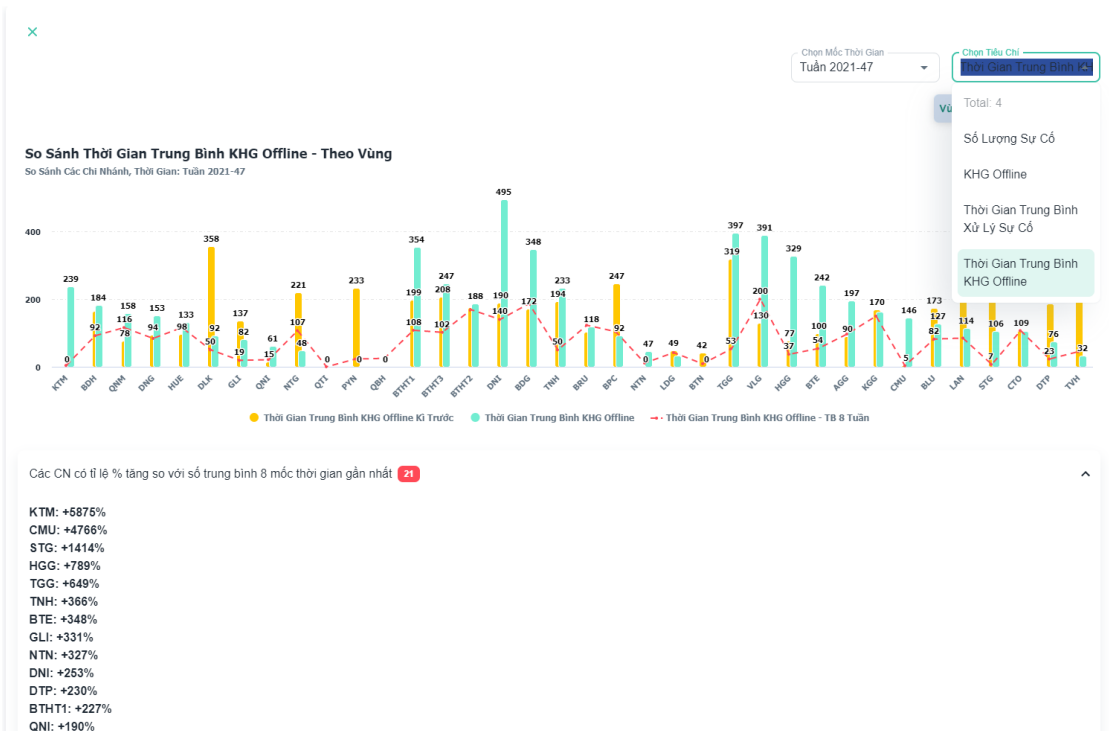


Figure 5.9: Comparison of average customer offline time

The chart above shows a comparison between the branches within the Southern Region, with options to view by Number of incidents, Offline customers, Average incident resolution time, Average customer offline time, or by time period.

The chart has two columns in two different colors, showing the figures for the selected time period and for the previous period, along with a line showing the 8-week moving average. Below the chart there are also statistics showing which branches have a percentage increase compared to the 8-week moving average.

Southern Region Incident Comparison by Error Detail

The final feature is a comparison of the number of incidents, the number of offline customers, the average incident resolution time, and the average customer offline time across the whole Southern Region, by error detail, filterable by time period.

These charts are visualized using bar charts, clearly showing the rise and fall in both counts and average times; they also include lines showing the upward/downward trend as well as the moving average of the data over time.

Users can choose to view statistics by error detail as well as by the corresponding time period.

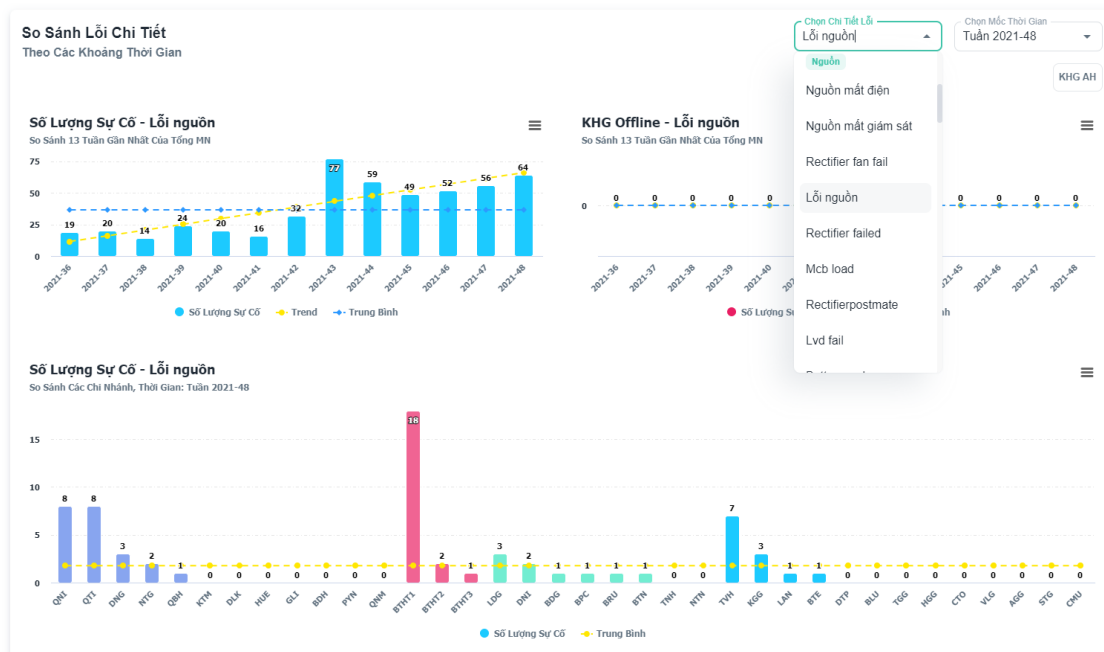


Figure 5.10: Southern Region incident comparison by error detail

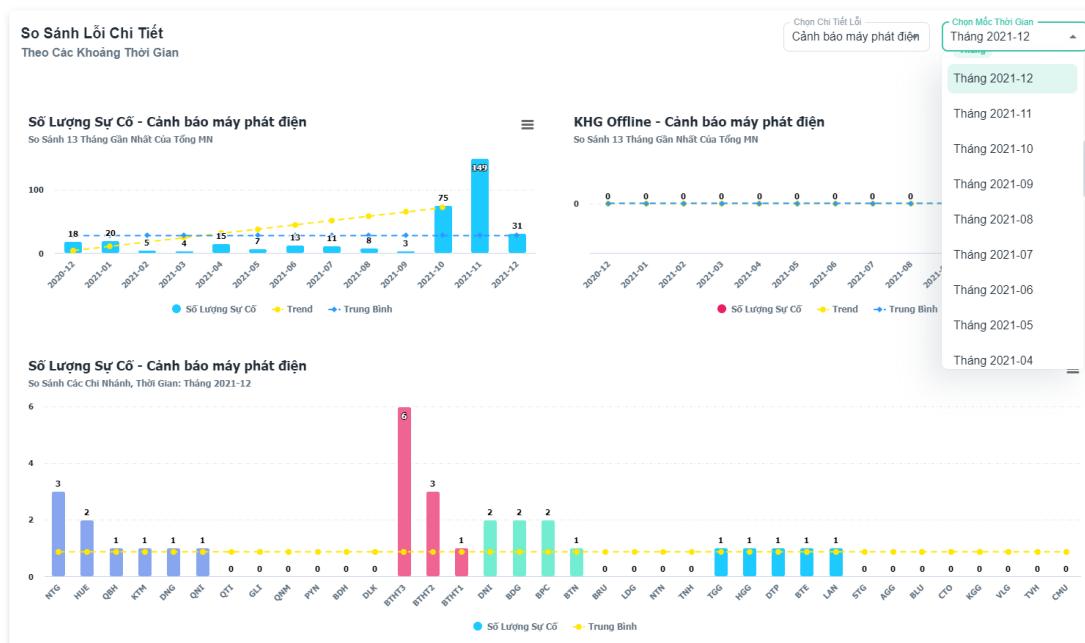


Figure 5.11: Southern Region incident comparison by time period

Dashboard Branch Page

To switch to the Dashboard Branch page, the user can select the Branch tab on the left side of the screen; the system then navigates to the Dashboard Branch page for infrastructure management by branch. This screen displays the reporting features and the options available for each. Below is the Dashboard Branch screen for the BTHT1 branch.

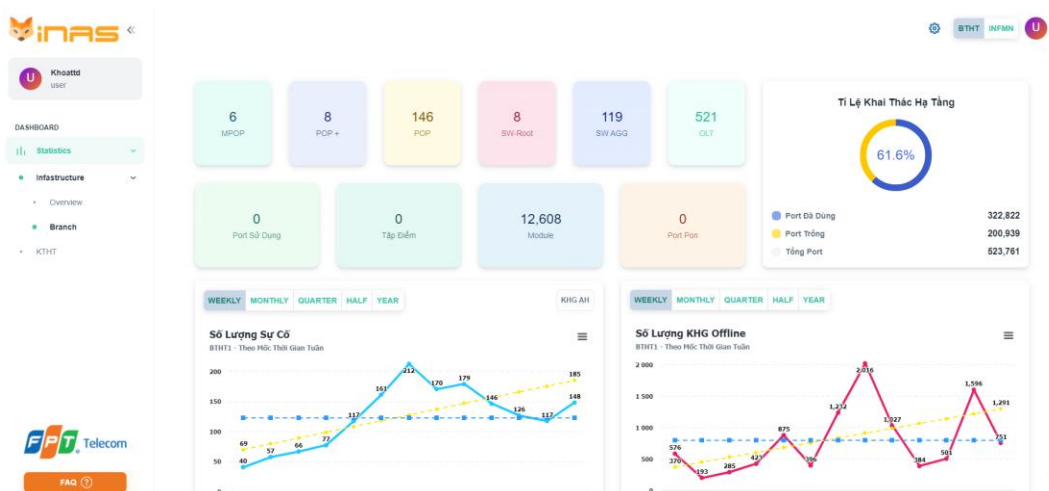


Figure 5.12: Dashboard Branch page

Branch Infrastructure Statistics

Below is the feature showing detailed infrastructure statistics for the BTHT1 branch. The screen shows the parameters for all devices belonging to the BTHT1 branch, along with the infrastructure utilization rate.



Figure 5.13: BTHT1 branch infrastructure statistics

The user can select any item to view a detailed list of information for it.

Danh Sách Module

Show Filter Columns Export

Tim Nhanh

ZONE	BRANCH	PROVINCE	POP	IP	DEVICE NAME	DEVICE TYPE	DEVICE MODEL	PORT	SERIAL NUMBER	PART NUMBER	SPEED	STATUS
V5	BTHT1	HCM	HCMP005	10.143.136.193	HCMP00502HW57	HUAWEI	HW57	GigabitEthernet0/0/0	1207L00899	GLC-BX-U	10B	Online
V5	BTHT1	HCM	HCMP005	10.143.136.193	HCMP00502HW57	HUAWEI	HW57	XGigabitEthernet0/0/3	20151208V2603	TS-SF-8512-02D-C	10B	Online
V5	BTHT1	HCM	HCMP102	10.143.121.89	HCMP10208DF24	DASAN	DF24	2	20160107W1696	TS-SF-BL3512-200	N/A	Online
V5	BTHT1	HCM	HCMP005	10.143.136.177	HCMP00505HW57	HUAWEI	HW57	XGigabitEthernet0/0/3	20150915W0030	TS-SF-8512-02D-D	10B	Online
V5	BTHT1	HCM	HCMP040	10.143.144.9	HCMP04003HW57	HUAWEI	HW57	XGigabitEthernet0/0/3	20160518W0418	GLC-BX10-D	10B	Online
V5	BTHT1	HCM	HCMP005	10.143.136.193	HCMP00502HW57	HUAWEI	HW57	GigabitEthernet0/0/6	1401L04968	GLC-BX-U	10B	Online
V5	BTHT1	HCM	HCMP102	10.143.121.89	HCMP10208DF24	DASAN	DF24	16	20150013W0979	TS-SF-BL3512-200	N/A	Online
V5	BTHT1	HCM	HCMP040	10.143.144.9	HCMP04003HW57	HUAWEI	HW57	GigabitEthernet0/0/5	HA13340540562	02315209	10B	Online
V5	BTHT1	HCM	HCMP005	10.143.136.177	HCMP00505HW57	HUAWEI	HW57	GigabitEthernet0/0/5	5071131000211	SFP-LX-10	10B	Online
V5	BTHT1	HCM	HCMP102	10.143.121.89	HCMP10208DF24	DASAN	DF24	24	1503L02527	TS-SF-BL3512-200	N/A	Online

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Figure 5.14: Detailed list of Modules under the BTHT1 branch

Danh Sách Switch AGG

Show Filter Columns Export

Tim Nhanh

PROVINCE	BRANCH	ARERA	PROVINCE	IPDEV	NAMEDEV	FUNCTION	GROUP	MPOP	POP	POPPLUS	POSINPOP
HCM	BTHT1	MN	V5	11.51.41.11	CE501555HCMP01801HW024	CE	ACCESS	01	P018	P555	01
HCM	BTHT1	MN	V5	11.51.41.201	CE501555HCMP24102HW024	CE	ACCESS	01	P241	P555	02
HCM	BTHT1	MN	V5	11.51.41.81	CE501555HCMP45001HW024	CE	ACCESS	01	P450	P555	01
HCM	BTHT1	MN	V5	11.51.61.121	CE501555HCMP00101HW067	CE	ACCESS	01	P001	P555	01
HCM	BTHT1	MN	V5	11.51.61.141	CE501555HCMP63901HW067	CE	ACCESS	01	P639	P555	01
HCM	BTHT1	MN	V5	11.51.61.31	CE501555HCMP44901HW067	CE	ACCESS	01	P449	P555	01
HCM	BTHT1	MN	V5	11.51.61.111	CE501555HCMP04201HW068	CE	ACCESS	01	P042	P555	01
HCM	BTHT1	MN	V5	11.51.62.171	CE501040HCMP23901HW068	CE	ACCESS	01	P239	P040	01

1 - 10 of 119 < 1 2 3 4 5 ... 12 > 10 / Page

Figure 5.15: Detailed list of AGG Switches under the BTHT1 branch

Branch Incident and Affected-Customer Statistics

The next feature on the Branch page is overview statistics on the number of incidents, the number of offline customers, the average incident resolution time, and the average customer offline duration for the BTHT1 branch over time, with the option to view statistics by Week, Month, Quarter, Half-Year, or Year.

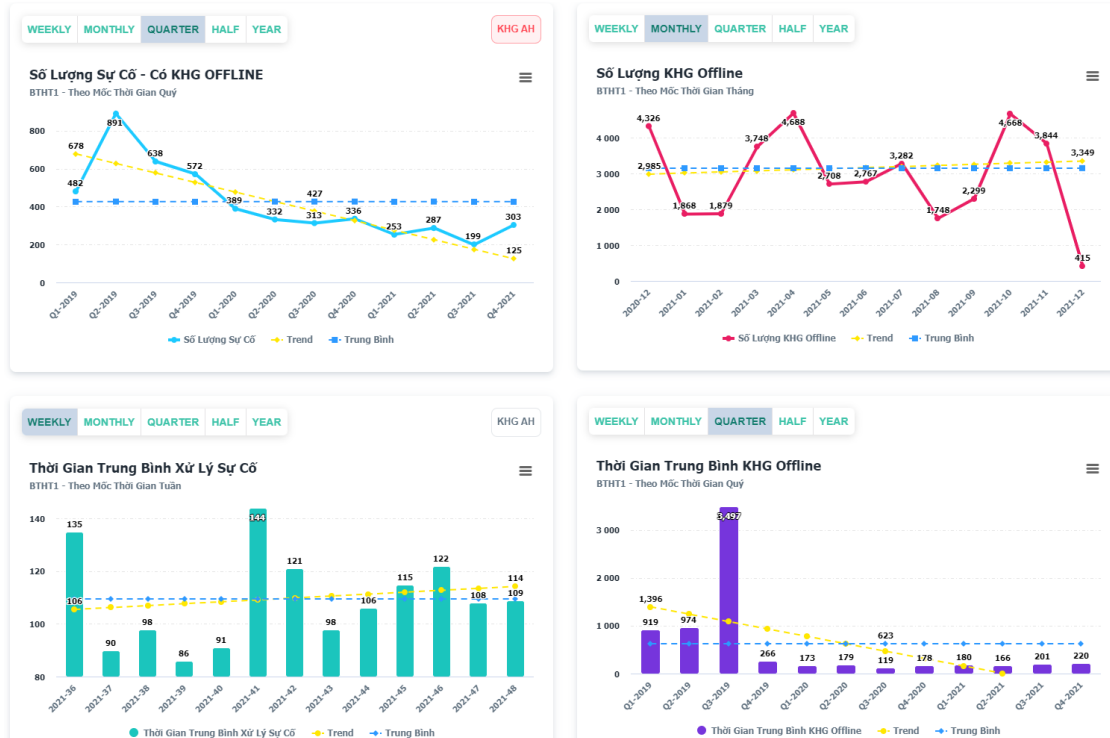


Figure 5.16: BTHT1 branch incident and affected-customer statistics by Week, Month, Quarter, Year, Half-Year

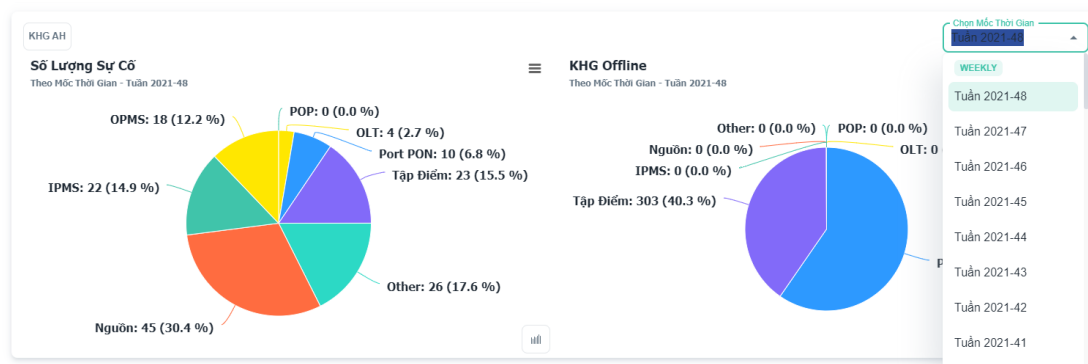


Figure 5.17: BTHT1 branch incident and affected-customer statistics by time period

Above is a pie chart showing the breakdown of the number of incidents and the number of offline customers by incident type, filterable by time period.

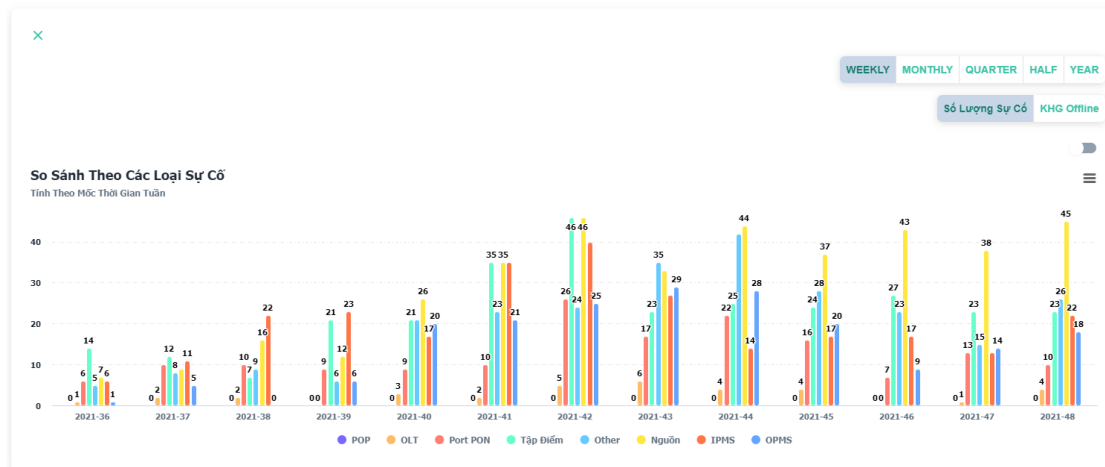
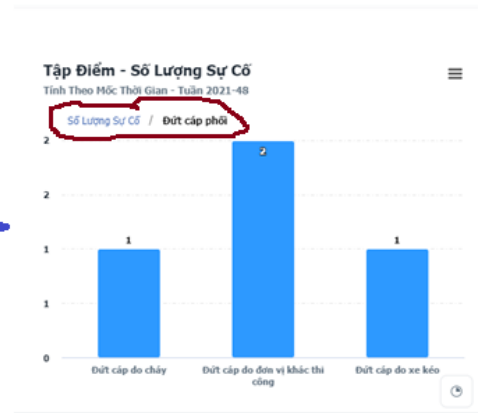
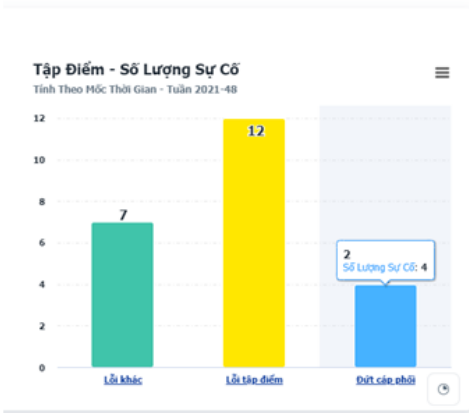


Figure 5.18: Comparison by incident type for the BTHT1 branch

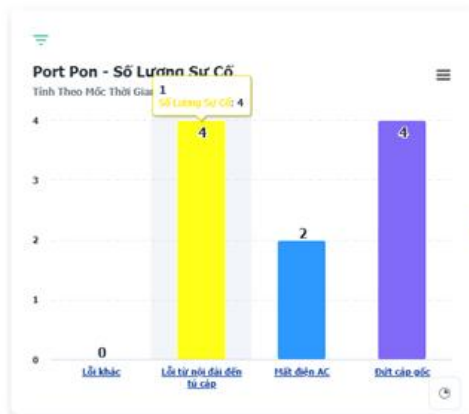
This is a chart comparing the overview number of each incident type, selectable by Incidents or Incidents with offline customers, and viewable by Week, Month, Quarter, Half-Year, or Year.



Figure 5.19: Detailed error comparison and top-5 statistics



The chart above shows the number of incidents as well as the number of offline customers over time, broken down by the detailed error for each incident type. It also includes a list of the top 5 POPs with the most offline customers, the top 5 incidents with the most offline customers, the incident with the longest resolution time, and overdue incidents.



The two charts above are two-level drilldown charts showing the number of Port-Pon incidents as well as access-point incidents, and the detailed error information for that incident type.

The chart uses Highcharts' HighchartDrilldown feature to show the detailed information behind a column in the main chart. When the user clicks on any column, the chart switches to a second level showing the statistics for the incident type behind the column the user clicked on the first-level chart.

System Engineering Dashboard Page

When the user selects the System Engineering (KTHT) Metrics, the system navigates to the System Engineering Dashboard page, the infrastructure management page for the System Engineering Department. This screen displays the reporting features and the options available for each.



Figure 5.22: System Engineering Dashboard page

Device Count Statistics

Below is the Device Count Statistics screen managed by the System Engineering Department, selectable by Total device count, Device type, Function, or Model.

First is a chart showing the total device count over time. This chart is a line chart across time periods, giving the user a visual sense of how the device count changes over time.

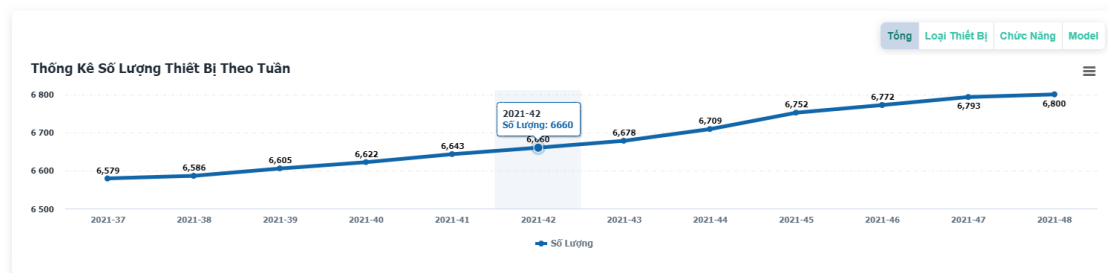


Figure 5.23: Total device count statistics by week

Next are charts showing device statistics by Type, by Function, and by Model. These charts are all shown as bar charts sorted in descending order of count, letting the user visually compare device counts by Type, Function, or Model as needed.

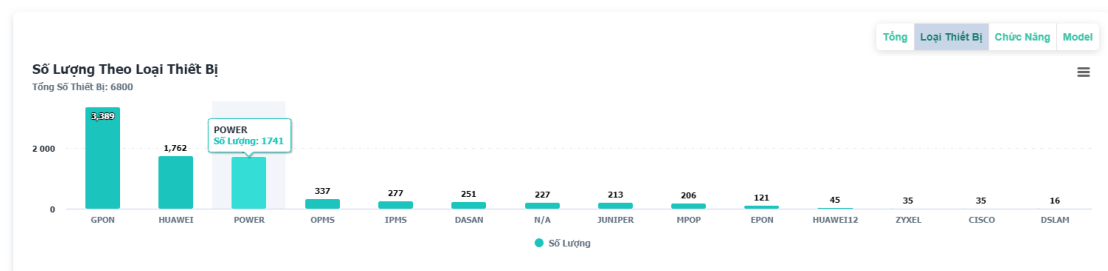


Figure 5.24: Device count statistics by type

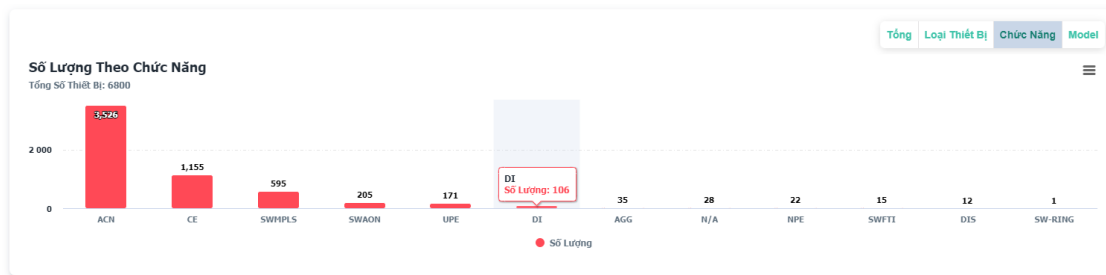


Figure 5.25: Device count statistics by function

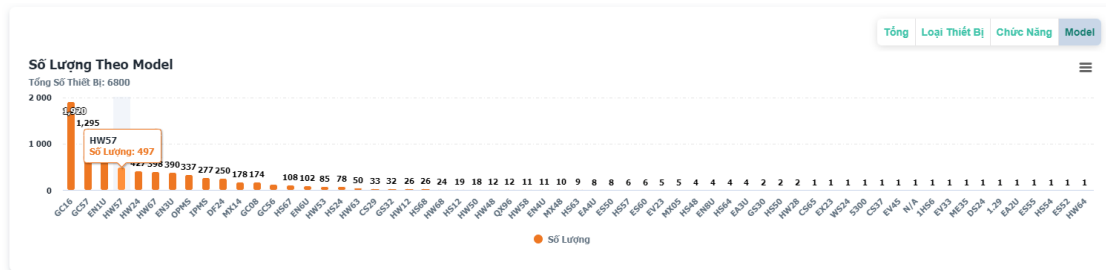


Figure 5.26: Device count statistics by model

INF Infrastructure Error Statistics

The feature providing detailed statistics on device errors and incidents within the System Engineering Department is presented as an overview through figures for each component category: OLT Uptime over 1 year, HW Uptime over 1 year, configuration errors, load-imbalanced ports, duplicate-OP devices, Port Unselect errors, and statistics for each error/incident category.

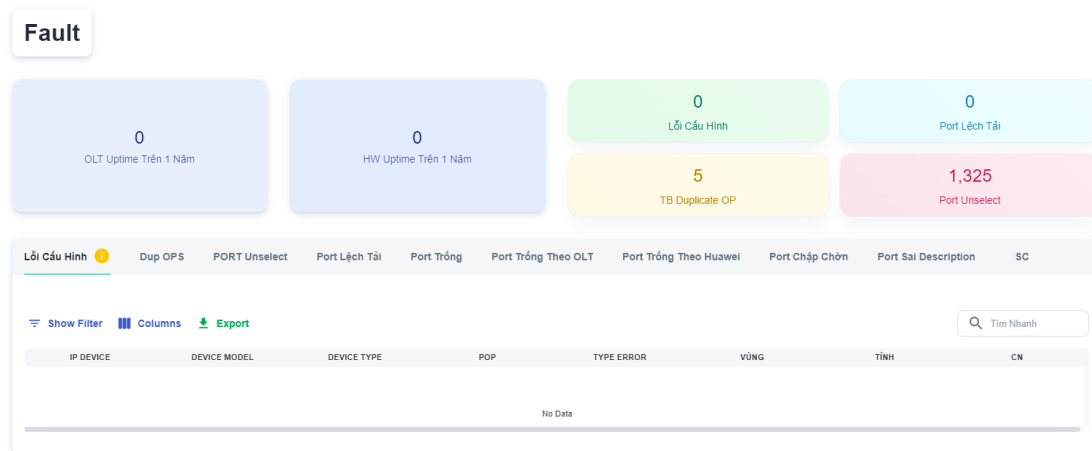


Figure 5.27: INF infrastructure error statistics

Since this is a detailed list, a table is used here to show the detailed list of information for each error and incident type.

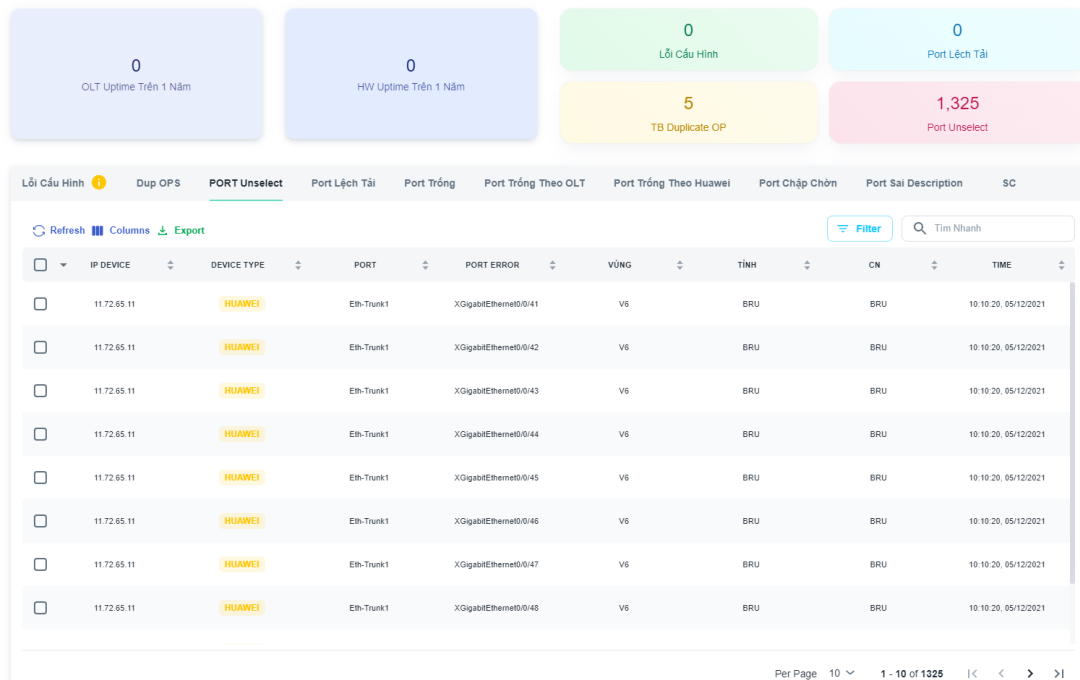


Figure 5.28: List of devices with Port Unselect errors

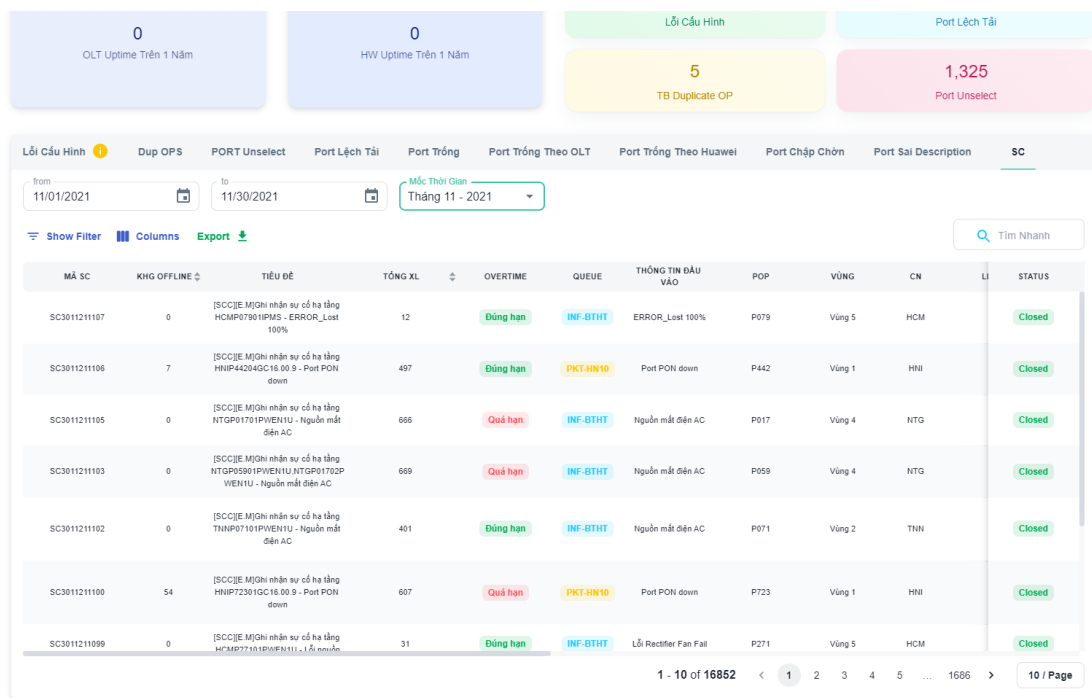


Figure 5.29: Incident list statistics

Device Health-Check Statistics

Below is the feature for statistics on device health-check information by severity level, the top 10 most frequently occurring log types, and the list of devices with recorded physical and service errors, filterable by various criteria.

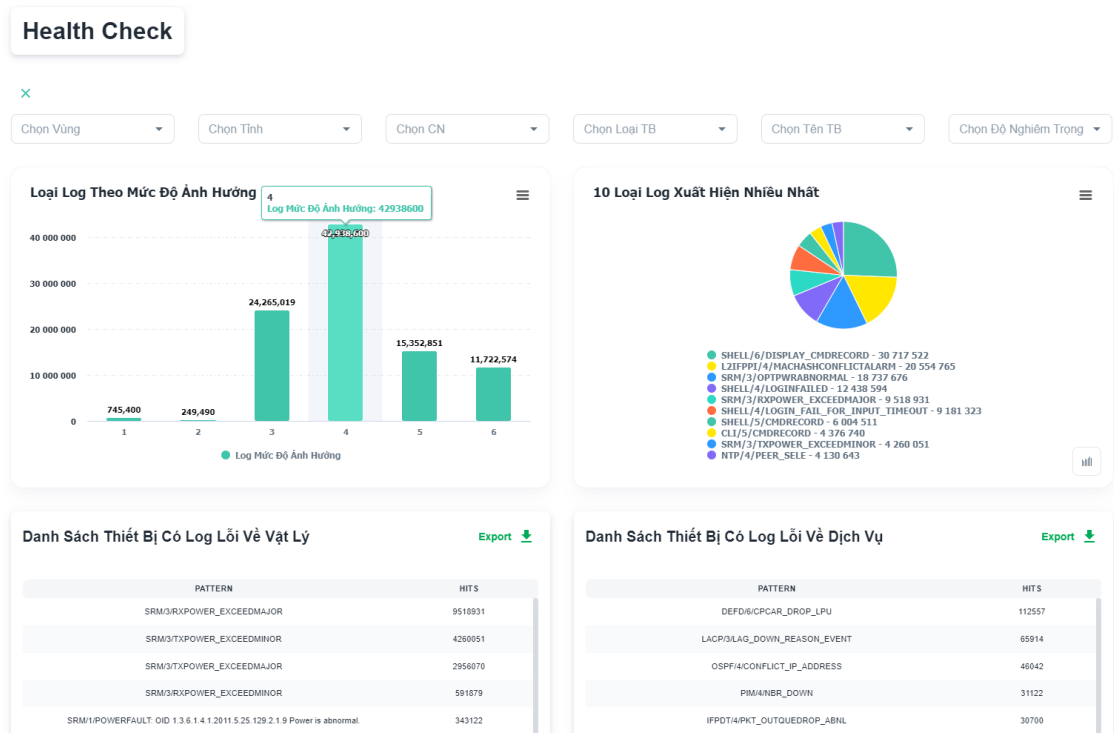


Figure 5.30: Device health-check statistics

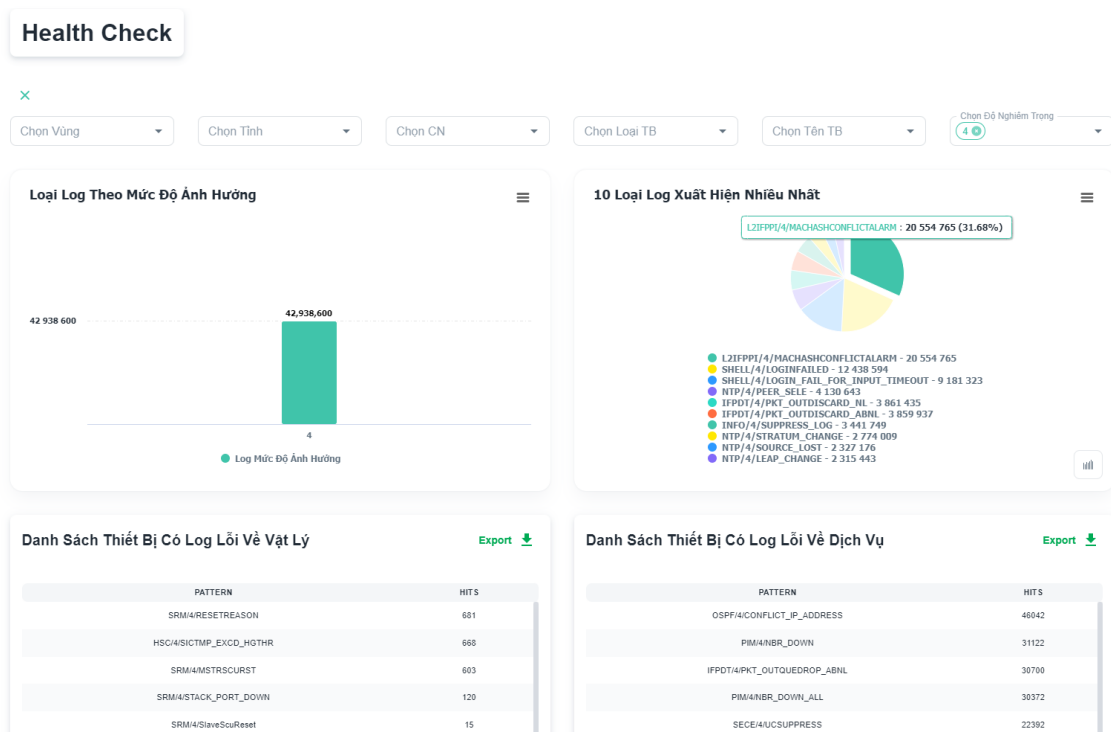


Figure 5.31: Log statistics by severity level

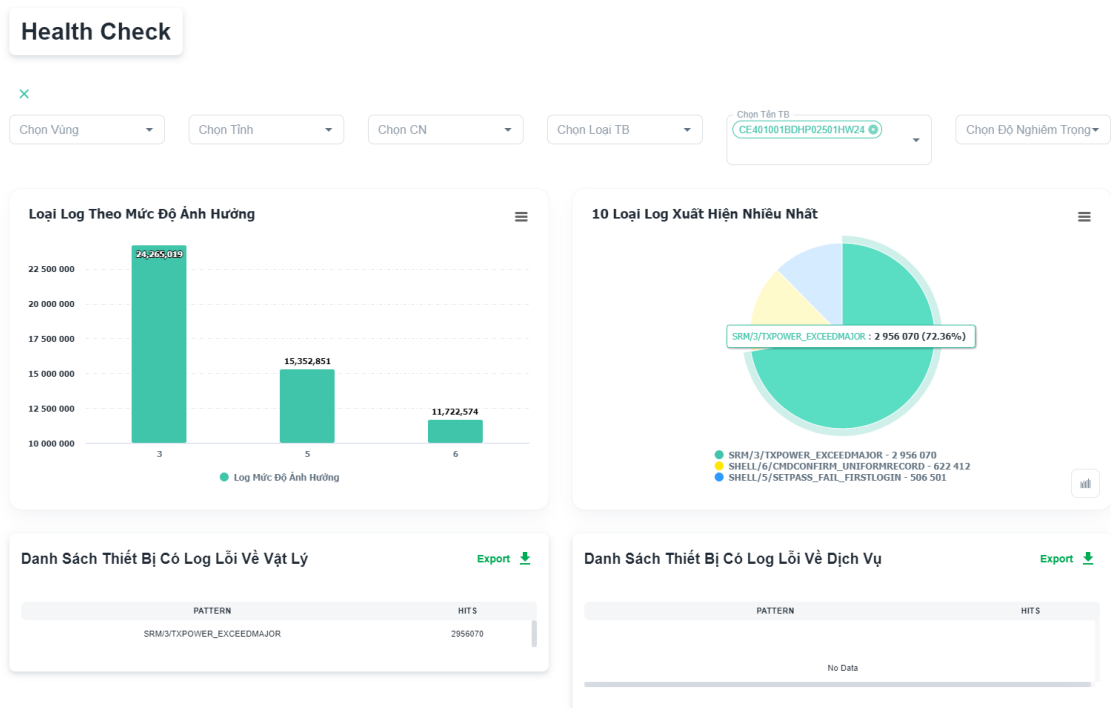


Figure 5.32: Log statistics by device name

System Engineering Department Incident-Handling KPI Statistics

The KPI statistics feature for average incident resolution time, support handling time, and the percentage of customers offline due to incidents for the System Engineering Department is shown as a bar chart, with lines showing the upward/downward trend as well as the moving average of the data over time.

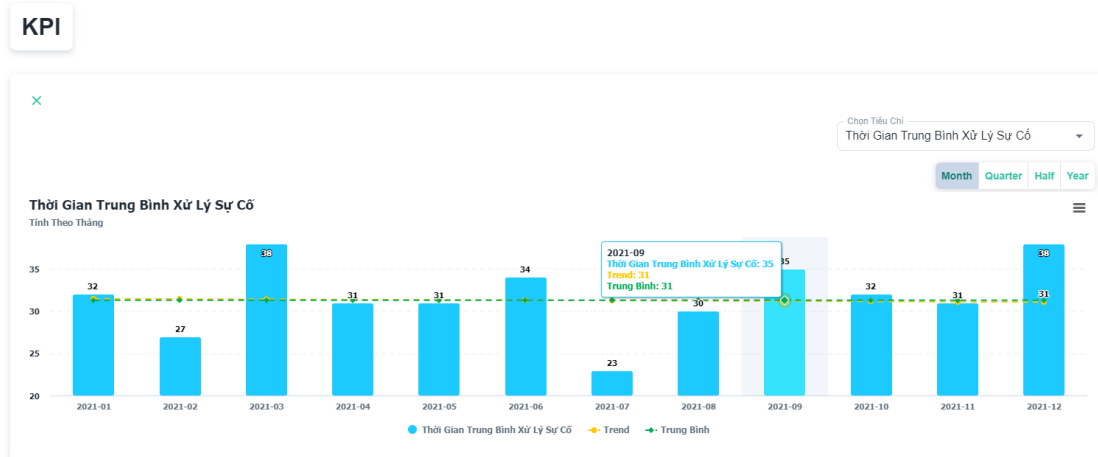


Figure 5.33: Average incident resolution time for the System Engineering Department

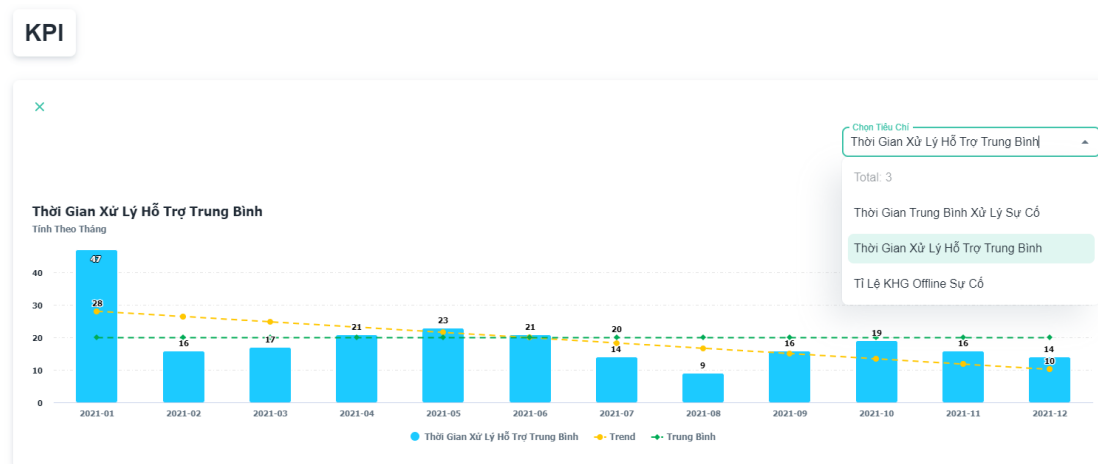


Figure 5.34: Average support handling time for the System Engineering Department

There are also statistics filterable by criteria as well as by Month, Quarter, Half-Year, or Year, letting the user view and assess how well the unit is meeting its targets.

KPI

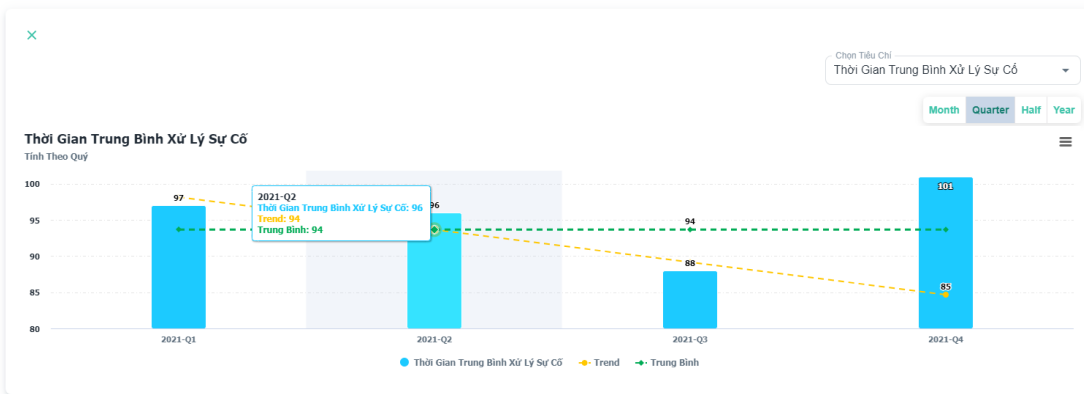


Figure 5.35: Average incident resolution time by quarter

System Engineering Incident Statistics

The final feature on the KTHT Dashboard page is incident statistics by type, detailed incident statistics, and support handling statistics for the System Engineering Department.

Sự Cố KTHT



Figure 5.36: System Engineering incident statistics



Figure 5.37: System Engineering incident statistics by criteria, time period, and region

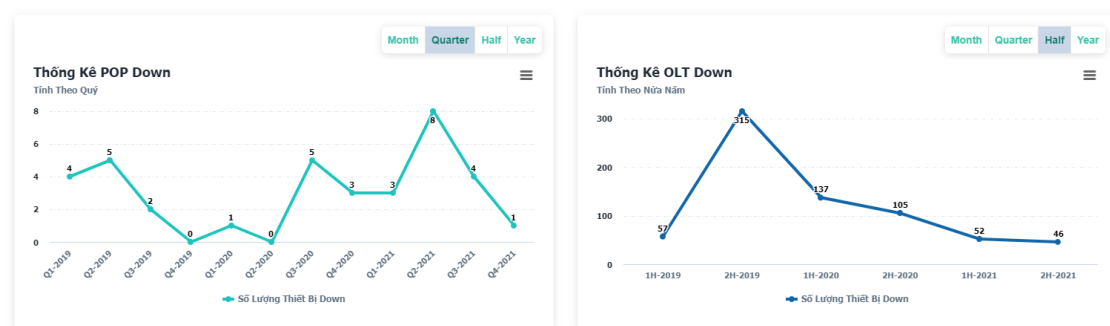


Figure 5.38: Pop Down and OLT Down statistics

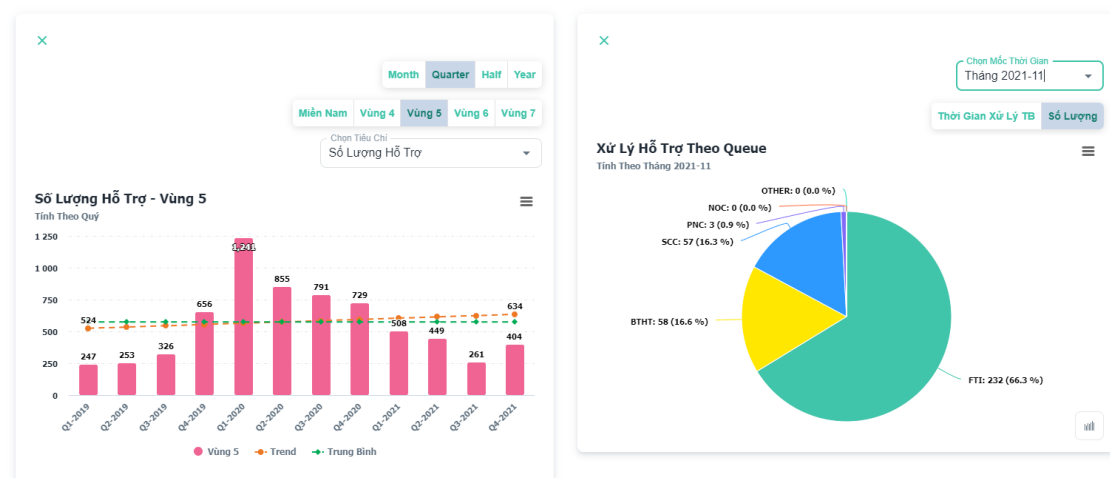


Figure 5.39: Support handling statistics by time period

System Evaluation

The system has been deployed and has achieved its stated goal of building a telecom infrastructure management system whose information and data are displayed in a visual, easy-to-understand way.

Below is a table assessing how complete the system's main features are:

No.	Feature	Completion Level
1	Southern Region Overview statistics	100%
2	Southern Region incident and affected-customer statistics	100%
3	Southern Region incident comparison by error detail	100%
4	Branch infrastructure statistics	100%
5	Branch incident and affected-customer statistics	100%
6	Device count statistics	100%
7	INF infrastructure error statistics	100%
8	Device health-check statistics	100%
9	System Engineering Department incident-handling KPI statistics	100%
10	System Engineering incident statistics	100%

Table 5.1: Assessment of feature completion in the system

Company Feedback After Deploying the Application in Production

After the system was built, the student received support from the System Engineering Department's Software Team in testing it and bringing it into real-world operation. After a period of use, the company was very pleased with the project and the features it provided.

The company gave feedback, which the student has summarized as follows:

- **On the interface:** the company rated the system's interface highly. The interface is simple, visually appealing, easy to use, and suitable for a wide range of users. Features are laid out and arranged sensibly on the interface. That said, the company also asked for further improvements to make the interface livelier, to display features more clearly, and to add notes/tooltips for buttons and options to make it easier for users to operate. This can be improved in the future, once the student has more time to research and develop the application's UI/UX to best fit its users.
- **On features:** the application is nearly complete and fully meets the business feature requirements the company originally set out. The application is organized around a main Dashboard Overview page – an overview of the telecom station system – along with component Dashboard pages broken down by the business needs of each department within the company, each with its corresponding features fully designed and built. When it first went live, the application had a few minor feature bugs; the

student took in the company's feedback and made fixes and improvements to complete the application.

- **On data visualization:** the application's data visualization was also rated quite highly. Compared to before, the statistical and comparative data is now far more visual. This makes it easier and more intuitive for the company to track management, report on data, and produce periodic reports, saving a great deal of time and effort. The company also suggested diversifying the chart types and adding more chart interactivity to make the system livelier.
- In addition, the company also intends to continue developing this project further – refining its existing features, building additional new features, and setting a long-term development direction for the project.

Chapter 6: Conclusion and Future Directions

This chapter summarizes the project's implementation process, the results achieved, and the future development directions for the project.

Conclusion

After researching, studying, and interning at the company, the student successfully carried out the project and achieved the goals set for it. Through this project, the student gained a great deal, both in personal knowledge and with respect to the project's system:

- The student was able to define the project's objectives and approach in order to carry it out quickly and completely.
- Thanks to the internship at the company, the student learned and greatly improved many workplace skills, such as: teamwork skills, interacting with other members of the company, risk management skills, managing and scheduling work over time, critical-thinking skills, independent research skills, and problem-solving and analytical thinking.
- The student completed the task of surveying, comparing, and selecting the technologies for the system. The student learned how to build a Frontend web application with the ReactJS framework NextJS based on the JavaScript programming language, and how to design a sensible interface, combined with the Highcharts and ApexCharts libraries to visualize data in line with the project's requirements.
- The student also learned how to build a Backend server with Express NodeJS combined with a MongoDB database, making it easy to deploy data into the system and meeting the data-retrieval and reporting requirements set out.
- Building on this knowledge, together with analysis and design skills, the student successfully built a system that meets both the business and technical requirements,

ensuring the system can operate well, meets users' needs, rarely runs into problems, and can be easily improved and maintained in the future.

Difficulties Encountered During the Project

During the project, the student also ran into a great many difficulties in the process of analyzing, researching, and deploying the project.

The difficulties and limitations include:

- Difficulty in system design analysis due to the old system structure, a lack of documentation, and a complex database.
- Difficulty when first learning and deploying technologies such as ReactJS and Express.js, and the architectures used to build web applications.
- Differences of opinion, and difficulty fully understanding the intent of company colleagues and the supervisor at times.
- Since the project was carried out solo, the workload was fairly heavy.
- The project timeline was short, so the application could not be fully completed within it.

Future Directions for the Project

Based on the results achieved, the student proposes the following directions for further development of the project:

- Add features beyond data visualization, such as device management (add, delete, edit) and searching for device information by managing area or region.
- Error/incident management, with automatic error alerts sent directly to the responsible handling staff for timely resolution.
- Visualize each device's location on a map using the Google Maps API.
- Manage information and statistics on storm/flood-affected areas in order to devise remediation, storm/flood-prevention solutions and reduce the impact on the infrastructure system.
- Improve performance, increasing data-retrieval speed as well as page-load speed.
- Improve the interface and user experience in the web browser. Consider and pursue development of a mobile application. Add more features to meet user needs and convenience.

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